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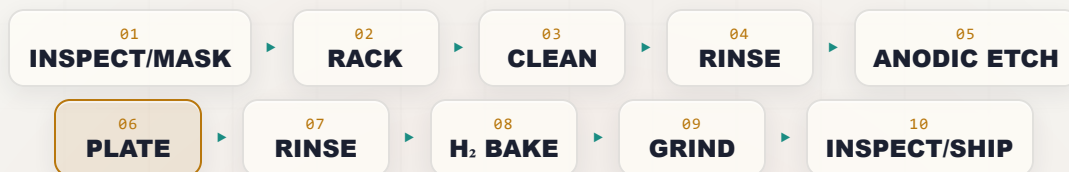
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PLATING POSTERS INC • METAL FINISHING REFERENCE SERIES

HARD CHROME

THE COMPLETE TRAINING MANUAL

A full training course for the brand-new operator — from “what is an anode?” to a signed certificate of completion. Functional (industrial) hard chromium from hexavalent chromic acid on steel. Assumes you know nothing. Teaches you everything — including the most dangerous chemistry on the whole line.



OPERATOR TRAINING & CERTIFICATION

EDITION 1.0 • 2026 • BEGINNER-FRIENDLY STYLE

Training reference only. **Hexavalent chromium (Cr VI) is a confirmed human carcinogen.** Always verify exact parameters against your chemistry supplier's Technical Data Sheets (TDS), customer specifications, and current Safety Data Sheets (SDS), and comply with OSHA 29 CFR 1910.1026 (Cr VI), EPA, REACH, and local regulations before production use.

WELCOME ABOARD

IF YOU DON'T KNOW AN ANODE FROM AN APRON, YOU'RE IN THE RIGHT PLACE — AND HERE THAT'S ABOUT YOUR HEALTH, NOT JUST QUALITY

Welcome. If you are holding this manual, you are about to learn how to run — or at least understand — a **hard chrome plating line**. Maybe you started this week. Maybe you've been racking parts for a year and finally want to know *why* you do the things you do. Either way, you're in the right place, and we're glad you're here.

Here is the most important thing we can tell you up front, and we'll say it more than once because it matters more than anything else in this book: **hard chrome plating uses hexavalent chromium — Cr(VI) — which is a confirmed human carcinogen**. It is the single most hazardous chemistry on the entire plating line. This is not said to scare you off. A trained, careful operator who respects this chemistry can work with it safely for an entire career — an untrained or careless one cannot. This manual is going to teach you that respect, station by station.

The second most important thing: **this manual assumes you know nothing about plating, chemistry, or electricity**. That's not an insult — it's a promise. Every term gets defined the first time we use it. Every step explains not just *what* to do but *why* it matters.

This manual is written in the spirit of the *plain-language* guides — friendly, plain-spoken, and patient. We would rather over-explain than leave you guessing in front of a tank full of chromic acid.

★ REMEMBER

You do not need to memorize this manual. You need to *understand* it. Understanding sticks. Memorizing fades by Friday — and on a Cr(VI) line, the understanding is what keeps you healthy.

• HOW TO USE THIS MANUAL

The manual follows the **line itself**, station by station, in the order a part actually travels: inspect/mask, rack, clean, rinse, etch/activate, plate, rinse, bake, grind to size, and inspect/ship. Reading it in order matters — the order of a plating line is never random, and neither is the order of this book.



TIP

Read the Fundamentals (Part One) all the way through *before* you read any individual station chapter. The station chapters assume you already understand the big picture — especially the Cr(VI) hazard — that Part One gives you. It's the difference between reading a recipe when you already know how to cook versus the very first time you've stepped into a kitchen.

THE ICONS — YOUR FRIENDLY MARGIN HELPERS



TIP

A shortcut, a trick of the trade, or a smarter way to do something — the kind of thing a 20-year veteran would lean over and tell you.



REMEMBER

A core idea worth locking in. If you forget everything else on the page, don't forget these. They're the load-bearing ideas.



HEADS-UP

A safety warning. Read every single one. On a hard chrome line these keep your lungs, skin, eyes, and long-term health intact. We use them very liberally — more than any other manual in this series — because Cr(VI) earns it.



TECHNICAL STUFF

Deeper chemistry or theory for the curious. Optional. Skip these and you can still run the line. Read them and you'll understand it on a deeper level.



FUN FACT

A bit of interesting trivia to make the science stick.

TWO MORE THINGS YOU'LL SEE AT EVERY STATION

Each station chapter ends with a **Teach-Back** checklist (paired with a "Top Mistakes" card) — the things you must say out loud, unprompted, before you're cleared — and a bordered **Sign-Off** box your *trainer initials* once you've demonstrated the skill safely. Treat the Teach-Back as your study guide and the Sign-Off as the gate you must pass.

— FIND YOUR WAY

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REMEMBER

This manual is a training and reference tool — **not** a substitute for your facility's written procedures, your supplier's TDS/SDS, your shop's OSHA Cr(VI) compliance program, or your supervisor's instructions. When this manual and your shop's official procedure disagree, **follow your shop's official procedure and ask your supervisor.**

WHAT YOU'LL BE ABLE TO DO

LEARNING OBJECTIVES

MASTER THIS AND EVERY STATION CHAPTER CLICKS INTO PLACE

By the time you finish this manual, you should be able to confidently:

1. **Explain what electroplating is** in plain language — how electricity moves metal onto a part.
2. **Describe what hard (functional) chrome does** and why a shop chooses it — extreme hardness, wear and abrasion resistance, low friction, corrosion resistance, and salvage/build-up of worn parts.
3. **Explain why hard chrome is “functional,” not “decorative”** — it's plated *thick* and usually *ground to a final dimension*, where decorative chrome is a flash over nickel.
4. **State the central safety fact of this line:** the bath is **hexavalent chromium (Cr VI)**, a **confirmed human carcinogen**, and name the controls that keep you safe (ventilation, mist suppression, respiratory protection, hygiene, no eating/smoking).
5. **Name the stations of the line in order** and explain *why* the order can't be rearranged.
6. **Explain the unusual electrochemistry** — very low cathode efficiency (~12–25%), very high current density, lots of hydrogen, and a microcracked deposit.
7. **Understand thickness control by ampere-hours** — the core operator math of this trade.
8. **Recognize hard chrome's biggest process challenge** — terrible throwing power — and the tools that fix it (conforming anodes, robbers/thieves, careful racking).
9. **Explain why a hydrogen-embrittlement bake is mandatory** on high-strength steel.
10. **Spot trouble early** and talk the language with your lead, your chemist, and your supplier.



REMEMBER

You are not expected to hit all of these on day one. Plating is a craft. But the *safety* objectives (#4) are not optional and not gradual — you must have those before you work the line at all.

• THE LINE AT A GLANCE

The whole functional hard chrome line in one strip. Don't memorize yet — feel the rhythm: **Inspect/Mask** → **Rack** → **Clean** → **Rinse** → **Anodic Etch** → **PLATE** → **Rinse** → **Bake** → **Grind** → **Inspect/Ship**.



REMEMBER

Notice the pattern: **a rinse follows the working tanks**, the **bake is not optional** on hard steel, and the part is usually **ground to final size after plating** — because hard chrome is plated thick and deliberately oversized. That last idea makes this line very different from the zinc family.



FUN FACT

Hard chrome and decorative chrome use almost the same bath chemistry — chromic acid and catalyst. The difference is *how much* you plate. Decorative “chrome” on a motorcycle is often only 0.25 µm thin over bright nickel; functional hard chrome can be hundreds of times thicker and is the real workhorse of industry.

CHAPTER 1

WHAT IS ELECTROPLATING?

THE TRUE BEGINNER'S PRIMER — SO FAR BACK WE DON'T EVEN ASSUME YOU KNOW WHAT PLATING IS

THE ONE-SENTENCE VERSION

Electroplating is using electricity to coat one metal with a layer of another metal. That's the whole idea. Now let's unpack it.

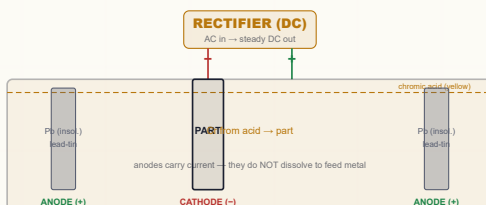
Imagine a worn hydraulic cylinder rod made of steel. It's strong, but its surface has worn down below the size it needs to be, and steel by itself scratches and corrodes. You want to give that rod a hard, smooth, wear-resistant skin of **chromium** — and even build the diameter *back up* to size. You can't paint that on. Instead, we use electricity to grow a tightly-bonded layer of chromium directly onto the steel, atom by atom. That's electroplating.

HOW ELECTRICITY MOVES METAL

Picture a tank of liquid called the **electrolyte** (or "the bath"). In hard chrome, that liquid is **chromic acid** — chromium dissolved in water in a charged form. You hook two things to a power supply called a **rectifier** (it converts ordinary AC wall power into the steady one-direction DC current plating needs):

- **The cathode** — your *part*, connected to the **negative** terminal. "*Cathode = the part, the thing collecting metal.*"
- **The anode** — bars or plates hanging in the same tank, connected to the **positive** terminal.

When you turn on the rectifier, positively-charged chromium species in the bath are attracted to the negative part, reach the surface, grab electrons, and turn into solid chromium metal stuck to the part. The longer you plate and the more current you push, the thicker the coating.



TECHNICAL STUFF

Here's the first big difference from zinc. In zinc plating the anodes are made of zinc and *dissolve to feed the bath*. In hard chrome, the anodes are **insoluble** — usually **lead, lead-tin, or lead-antimony alloy** (or, in some shops, platinized or mixed-metal-oxide types). They do **not** dissolve to supply chromium. Instead, all the chromium comes from the **chromic acid you add to the bath**, and the anodes' main job is just to carry current and keep the chemistry oxidized. So unlike a zinc tank, **a hard chrome tank is not self-feeding for metal — you replenish chromium as a chemical, not as a dissolving anode.**

REMEMBER

Three words to lock in: **Cathode = the part** (collects chrome). **Anode = insoluble lead** (carries current, does *not* dissolve to feed metal). **Electrolyte = chromic acid** (supplies the chromium). Almost everything on this line is a variation on this one idea.

HEADS-UP

Plating rectifiers push enormous current — hard chrome runs at **very high amperage** (often thousands of amps on a big tank). High current, low voltage, into a tank of hot, corrosive, conductive acid. **Never reach into an energized tank, and never do maintenance until the rectifier is locked and tagged out (LOTO).**

CHAPTER 2

WHAT IS “HARD CHROME”?

FUNCTIONAL CHROMIUM — HARDNESS, WEAR RESISTANCE, LOW FRICTION, AND BRINGING WORN PARTS BACK TO LIFE

CHROME PLATING COMES IN TWO VERY DIFFERENT JOBS

“Chrome plating” can mean two completely different things:

- **Decorative chrome** — an *extremely thin* flash of chromium (typically **0.13–0.5 µm**) over a base of bright nickel, used for shine and tarnish resistance on car trim, faucets, motorcycle parts. The nickel underneath does most of the corrosion work; the chrome is mostly cosmetic.
- **Hard (functional/industrial) chrome** — what *this* manual is about. A **thick, functional** chromium deposit plated **directly on steel** (no nickel underneath), valued not for looks but for **performance**: extreme hardness, wear and abrasion resistance, low friction, and the ability to **rebuild worn parts back to size**.

★

REMEMBER
Hard chrome is a *functional* deposit, not a *decorative* one. It is plated *thick* and almost always *ground to a final dimension* afterward. If you remember one sentence about what makes this line different from the zinc family, make it that one.

WHAT HARD CHROME ACTUALLY GIVES A PART

PROPERTY	WHAT IT MEANS ON THE FLOOR
Hardness	Typically 800–1100 HV (Vickers) — among the hardest platable coatings. Resists scratching, galling, and wear.
Wear / abrasion resistance	Surfaces that slide, rub, or take impact last far longer.
Low friction	Chrome’s low coefficient of friction means smoother sliding and less heat in moving parts.
Corrosion resistance	Resists many chemicals and atmospheric corrosion (though the microcrack network limits this).
Build-up / salvage	Plate worn or undersized parts <i>oversize</i> , then grind back to spec — saving expensive parts instead of scrapping them.
Release / anti-stick	Used on molds and dies so plastic and rubber release cleanly.

Where you’ll see it: hydraulic cylinder rods and pistons, shock-absorber rods, engine valve stems, piston rings, printing rolls, plastic-injection molds, dies, gun bores, shafts, gauges, and any part that must survive heavy wear and be held to a precise dimension.

💡

FUN FACT
Hard chrome’s hardness comes partly from how it grows: the deposit forms with a built-in network of tiny cracks and very fine crystal grains. The same microcrack structure that makes it hard is also why it isn’t a perfect corrosion barrier on its own — moisture can find its way down the cracks to the steel.

WHAT “THICK” MEANS HERE

Decorative chrome is a fraction of a micron. **Functional hard chrome runs from a few microns to several hundred microns** (and in heavy build-up/salvage work, even thicker).

APPLICATION	TYPICAL CHROME THICKNESS
Light wear / “thin dense chrome”	2–25 µm
General functional / hydraulic rods	20–100 µm
Heavy wear, salvage, build-up	100 µm to several hundred µm

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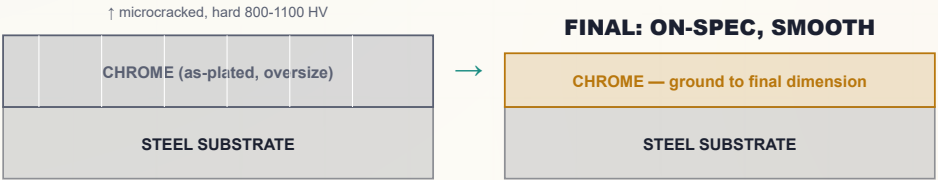
TECHNICAL STUFF
Because hard chrome is plated thick *and* the bath is slow (low efficiency — Chapter 6), plating times are long. At a typical deposition rate of roughly **25 µm/hr**, a 100 µm build takes around 4 hours of plating; a heavy salvage job can run a full shift or more. This is one reason ampere-hour control (Part Three) is such a core operator skill — you’re managing a long, deliberate process, not a quick dip.

THE MICROCRACKED STRUCTURE

A good hard chrome deposit is **microcracked** — covered in an extremely fine network of cracks, typically hundreds of cracks per linear centimeter. This is normal and, for wear performance, desirable: the cracks hold lubricant and relieve internal stress so the thick deposit doesn't peel. Crack pattern is actually a *quality check* (Part Three) — too few cracks, or a coarse “macrocrack” pattern, can mean a process problem.

PLATE OVERSIZE → GRIND TO SIZE (NOT TO SCALE)

FUNCTIONAL CHROME IS PLATED THICK, THEN MACHINED TO SIZE



REMEMBER

A microcracked surface is **normal and wanted** in hard chrome. Don't panic when you learn the deposit is “cracked” — that fine crack network is part of how it works. *Macrocracking* (big, coarse cracks) is the defect to watch for.

A WORD ON THE TRIVALENT ALTERNATIVE

A newer chemistry, **trivalent hard chrome — Cr(III)** — is emerging as a less-hazardous alternative because it avoids the carcinogenic Cr(VI). It's growing for *decorative* and thinner functional work, but for **thick, high-hardness functional deposits the hexavalent Cr(VI) process still dominates** — it remains the workhorse and is what this manual trains you on. Wherever this manual says “the bath,” it means the hexavalent chromic-acid bath unless it explicitly says trivalent.



HEADS-UP

Knowing that a trivalent alternative *exists* does **not** change how you must treat the hexavalent bath in front of you. The tank you run is Cr(VI), a carcinogen. Treat it that way, every shift.

CHAPTER 3 • THE MOST IMPORTANT CHAPTER

WHY HEXAVALENT?

THE CR(VI) HAZARD, UP FRONT — READ THIS CHAPTER TWICE

We put this chapter near the front on purpose. On every other line in this series, the safety chapter can wait until after you understand the chemistry. **Not here.** The defining feature of hard chrome plating is that it runs on **hexavalent chromium, Cr(VI)** — and Cr(VI) is a **confirmed human carcinogen**. You need to understand this *before* you read another word about how to plate.

WHY THE BATH HAS TO BE HEXAVALENT

The functional hard chrome bath is **chromic acid** — chromium trioxide (CrO_3) dissolved in water. When CrO_3 dissolves, the chromium is in its **+6 (“hexavalent”) oxidation state**. That hexavalent chromium, with a small amount of catalyst, is what makes it possible to deposit a thick, hard chromium layer at all. The same property that makes the chemistry work is what makes it dangerous.

 **TECHNICAL STUFF**

“Hexavalent” (Cr^{6+} , Cr VI) and “trivalent” (Cr^{3+} , Cr III) refer to how many electrons the chromium atom is missing — its oxidation state. Cr(VI) is a strong oxidizer and is highly toxic and carcinogenic. Cr(III) — the form chromium ends up in *after* it’s been used or reduced — is far less hazardous and is actually a trace nutrient. **The whole goal of waste treatment on this line (Part Three) is to convert the dangerous Cr^{6+} into the safe Cr^{3+} and remove it.**

HOW CR(VI) HURTS PEOPLE

This is not abstract. Cr(VI) exposure on a plating line causes real, documented harm:

- **Lung cancer** — the most serious. Long-term inhalation of chromic acid mist is linked to lung and sinonasal cancer. This is why the bath is regulated as a carcinogen.
- **Nasal septum ulceration and perforation** — chromic acid mist literally eats a hole through the cartilage wall between the nostrils. This was once a common, visible mark of long-time chrome platers. It is **preventable** with proper controls.
- **“Chrome holes” / chrome ulcers** — deep, slow-healing skin ulcers (often on hands, knuckles, forearms) where chromic acid contacts even a small break in the skin.
- **Skin and respiratory irritation, dermatitis, and allergic sensitization.**
- **Eye damage** — chromic acid is corrosive to the eyes.

 **HEADS-UP — THE SINGLE MOST DANGEROUS CHEMISTRY ON THIS LINE**

The hazard is mostly invisible. The chromic acid **mist/aerosol** thrown up off the tank by gassing and agitation is the main route into your lungs, and you can be over-exposed without seeing or strongly smelling anything. **Engineering controls (ventilation + mist suppression) are your primary protection — not just PPE.** If the ventilation or fume suppressant is not working, the tank is not safe to run. Stop and report it.

THE LAW: OSHA'S CR(VI) STANDARD

Cr(VI) has its own dedicated OSHA standard — **29 CFR 1910.1026**. You don’t need to memorize the legal text, but you must know the numbers your shop’s program is built around:

TERM	VALUE	PLAIN MEANING
PEL (Permissible Exposure Limit)	5 $\mu\text{g}/\text{m}^3$	The legal ceiling for Cr(VI) in the air you breathe over an 8-hr shift
Action Level	2.5 $\mu\text{g}/\text{m}^3$	The “pay attention” trigger (8-hr avg) — at or above this, the shop must do monitoring, training, and medical surveillance

A microgram (μg) is a *millionth* of a gram. A limit of 5 μg per cubic meter of air is *extraordinarily* small — which tells you how potent this hazard is and why mist control is non-negotiable.

**TECHNICAL STUFF**

Because the PEL is so low, you cannot judge your exposure by smell or eye irritation — those happen at far higher levels. That's why OSHA requires *air monitoring* (measuring the actual Cr(VI) in the air), *medical surveillance* (periodic health checks for exposed workers), regulated areas, a written exposure control plan, and specific hygiene facilities. Your shop's Cr(VI) program is a legal requirement, not a suggestion.

• THE FIVE PILLARS OF CR(VI) CONTROL

You'll see these at every station:

1. **Mist suppression in the tank** — a **fume suppressant / foam blanket** (a surfactant additive that makes a foam layer on the bath surface and lowers surface tension so bursting gas bubbles throw far less mist), and/or floating plastic balls. This is your first line of defense, right at the source.
2. **Tank ventilation** — **local exhaust** such as **push-pull** or lateral-exhaust hoods that pull mist off the tank surface, and often a **fume scrubber** that washes Cr(VI) out of the exhausted air before it leaves the building.
3. **Respiratory protection** — the right respirator for the task, per your shop's program, used *in addition to* (never instead of) ventilation.
4. **Hygiene and no hand-to-mouth** — **no eating, drinking, smoking, vaping, or cosmetics** anywhere near the line; designated wash-up; separate work clothing; decontamination before breaks and before leaving.
5. **Spill, waste, and skin protection** — full chemical PPE, immediate flushing of any skin/eye contact, controlled spill response, and proper **Cr(VI) waste treatment** (reduce $\text{Cr}^{6+} \rightarrow \text{Cr}^{3+}$, then precipitate — Part Three).

**REMEMBER**

Engineering controls first (suppress the mist, ventilate the tank), respirator second, hygiene always. PPE is the *last* layer of defense, not the first. A respirator does not make a tank with broken ventilation safe.

**FUN FACT**

The yellow-green color of the chromic acid bath comes from the hexavalent chromium itself. When waste treatment reduces it to trivalent, the color shifts toward green/blue — operators can sometimes *see* the reduction reaction happening as the color changes. (Color is a clue, never a substitute for proper testing.)

**HEADS-UP — HYGIENE IS PART OF THE JOB, NOT OPTIONAL**

Hand-to-mouth contact is a real route for getting Cr(VI) into your body. **Never** eat, drink, smoke, vape, or touch your face on the line. Wash thoroughly before every break and before going home. Keep work clothing and street clothing separate so you don't carry chromium home to your family.

**REMEMBER**

If you take only one thing from this chapter: the chromic-acid bath is a **confirmed human carcinogen** whose worst hazard is an *invisible mist*. Suppress the mist, ventilate the tank, wear your respirator, keep your hands away from your face, and never let Cr(VI) reach a drain. Do those things, every shift, and you can work this line safely for a career.

CHAPTER 4

HOW A HARD CHROME LINE WORKS

A LINE IS A SYSTEM, NOT A PILE OF SEPARATE TANKS

A hard chrome line is a **sequence of stations** a part travels through in a strict order. Each station does one job and prepares the surface for the next. Parts are almost always **rack plated** here — individually hung on a current-carrying **rack** — because hard chrome parts are usually large, precision, high-value items (rods, shafts, rolls, molds) that need careful positioning and often custom anodes. (Barrel plating, common in zinc, is rare in functional hard chrome.)

• THE WHOLE HARD CHROME SEQUENCE

#	STATION	THE ONE THING IT DOES	TEMP	NOTES
01	Inspect / Mask / Rack	Protect non-plated areas; position for current	Ambient	Masking is a chrome specialty
02	Clean / Degrease	Remove all oil, grease, shop soil	60–82°C	Alkaline soak/electroclean
03	Rinse	Wash off cleaner (the firewall)	Ambient	—
04	Anodic Etch / Activate	Reverse-current etch to “wake up” steel	~50–60°C	Often done <i>in the chrome tank</i>
05	Hard Chrome Plate	Deposit the thick, hard chrome layer	~50–60°C	Main event; Cr(VI)
06	Drag-Out / Rinse	Recover dragged-out chrome; wash the part	Ambient	Recovery matters — costly & hazardous
07	H-Embrittlement Bake	Drive absorbed hydrogen out of the steel	~190°C	Mandatory on high-strength steel
08	Grind / Finish to Size	Machine the oversize deposit to final dim.	—	Defines hard chrome as functional
09	Strip / Rework (as needed)	Remove bad chrome and re-plate	—	Reverse-current or chemical strip
10	Dry / Final Inspection	Dry and verify thickness, hardness, finish	—	Ship

**HEADS-UP**

Notice the bath temperature: hard chrome runs **hot, around 50–60°C (122–140°F)**, much hotter than a room-temperature acid zinc bath. A hot chromic-acid tank throws *more* mist, so ventilation and mist suppression matter even more. Hot + carcinogenic + corrosive = treat this tank with total respect.

• A LINE IS A SYSTEM, NOT A PILE OF SEPARATE TANKS

**REMEMBER**

Every station either **protects the next** or **poisons it**. There is **no neutral step**. A part that leaves the cleaner still oily contaminates the etch and won't bond. A part that skips the bake can crack in service. A part ground without checking thickness can end up undersize and scrap. What you do (or fail to do) early shows up as a defect — or a safety failure — later.

**TIP**

Think of the chrome tank as the crown jewel *and* the dragon of the line. Everything *before* it (inspect, mask, rack, clean, rinse, etch) exists to make sure only a perfectly clean, well-masked, properly racked part enters it. Everything *after* it (rinse, bake, grind, inspect) exists to make the chrome safe, sound, and exactly the right size. The whole line orbits that one hot, yellow, hazardous tank.

CHAPTER 5

WHY THE ORDER MATTERS

EACH STATION SETS UP THE NEXT, AND THE SEQUENCE IS LOCKED BY CHEMISTRY — AND BY SAFETY

INSPECT / MASK / RACK FIRST

Because hard chrome has terrible throwing power and you only want chrome where the customer wants it — masking and rack position decide where the metal goes *before* any chemistry happens.

CLEAN BEFORE ETCH

Because chrome, acid, and etch all need *bare metal*. Plate over oil and you've plated over oil — the chrome bonds to grease, not steel, and flakes off. The free, foolproof **water-break test** (clean steel sheets water in an unbroken film; dirty steel beads it) tells you when cleaning is done.

RINSE BETWEEN CLEANER AND ETCH

Because the alkaline cleaner and the acidic etch are chemical opposites; carry one into the other and you waste chemistry and weaken the etch.

ANODIC ETCH RIGHT BEFORE PLATING

Because freshly activated steel re-oxidizes in minutes. The anodic (reverse-current) etch removes the invisible oxide *and* a tiny bit of base metal to leave a chemically fresh, microscopically roughened surface for a true metallurgical bond. In hard chrome this is often done **in the chrome tank itself**, running the part as the *anode* (reverse polarity) for a short time, then switching to cathode to plate — a slick way to go straight from activation into plating with no exposure to air in between.



REMEMBER

No activation, no adhesion. Period. And on a thick, stressed hard-chrome deposit, poor adhesion doesn't just look bad — a hundred-micron chrome layer that peels can fail catastrophically on a precision part. Activation has to come right before plating.

PLATE, THEN RINSE, THEN BAKE, THEN GRIND, THEN INSPECT

In that order, because:

- you must **rinse** to recover dragged-out chrome and stop the carcinogen from spreading;
- you must **bake within hours of plating** to drive out hydrogen *before* it can crack a high-strength part;
- you **grind after baking** so the part is at its final, stress-relieved condition when you bring it to size;
- you **inspect last** to confirm thickness, hardness, and finish before it ships.



HEADS-UP

The bake comes **before** final grinding for a reason: you want the hydrogen driven out and the part metallurgically settled before you machine it to its final critical dimension. And the bake must start **soon after plating** (typically within a few hours — see Part Three) — a part left overnight before baking may already have cracked. Don't let plated high-strength parts sit.



TECHNICAL STUFF

The recurring villain that justifies all the rinses is **drag-out** (the film of solution clinging to a part as it leaves a tank) and its mirror, **drag-in** (that film contaminating the next tank). On a Cr(VI) line drag-out is doubly serious: it wastes expensive chromic acid *and* spreads a carcinogen around the shop and into the wastewater. That's why hard chrome lines use **drag-out recovery rinses** (Station 06) that capture chrome for return to the tank.

THE ELECTROCHEMISTRY OF HARD CHROME

LOW EFFICIENCY, HIGH CURRENT, LOTS OF HYDROGEN, AND A CRACKED DEPOSIT — WHY THIS BATH BEHAVES NOTHING LIKE ZINC

Hard chrome is electrochemically strange. Understanding *why* makes everything downstream make sense.

1. VERY LOW CATHODE EFFICIENCY

Cathode efficiency is the percentage of your electrical current that actually deposits metal (versus being wasted). Acid zinc runs ~95–98% efficient. **Hard chrome runs only about 12–25% efficient** — most of your current is “wasted” splitting water into **hydrogen gas** at the part and **oxygen** at the anode.



REMEMBER

Hard chrome is a low-efficiency, slow, current-hungry process. Only roughly **one-fifth** of the electricity you pay for becomes chrome. The rest makes gas. This single fact explains the high current, the long plating times, the heavy hydrogen, and why the tank needs serious cooling and ventilation.

2. VERY HIGH CURRENT DENSITY

Because efficiency is low and you need a lot of metal, you drive the part *hard*. **Cathode current density runs ~30–60+ A/dm² (~300–600+ ASF)** — roughly *ten times* the current density of a zinc bath. (“Current density” = amps spread over the part’s surface area; it’s amps *per unit area*, not total amps.)

PARAMETER	HARD CHROME (TYPICAL)	ACID ZINC (FOR CONTRAST)
Cathode efficiency	~12–25%	~95–98%
Cathode current density	~30–60+ A/dm ² (300–600+ ASF)	~1–6 A/dm ²
Bath temperature	~50–60°C	~21–35°C
Deposition rate	~25 µm/hr (typical)	much faster per A·h
Anodes	Insoluble (lead/lead-tin)	Soluble (zinc, feeds bath)

3. LOTS OF HYDROGEN → EMBRITTLEMENT RISK

All that “wasted” current makes **atomic hydrogen** at the part — and some of it diffuses *into* the steel. On high-strength/hardened steel this causes **hydrogen embrittlement**: trapped hydrogen makes the part brittle so it can **crack and fail suddenly under load**, hours or days after going into service. This is why a **post-plate bake is mandatory** on high-strength steel (Station 07 / Part Three).



HEADS-UP

The low efficiency that makes hard chrome slow is the *same* thing that makes it generate a lot of hydrogen. More hydrogen at the part means more soaks into the steel. On a high-strength rod or shaft, skipping the bake can turn a good part into a hidden time bomb. The bake is not paperwork — it prevents catastrophic failures.

4. THE MICROCRACKED DEPOSIT

Chrome plates with very high **internal stress**, and it relieves that stress by **cracking** — forming the fine microcrack network you read about in Chapter 2. Controlled microcracking is normal and desirable for wear. Process drift can produce the wrong crack pattern (too few cracks, or coarse macrocracks), which is a quality flag (Part Three).

5. THE BATH: CHROMIC ACID + CATALYST

The bath is **chromic acid** (CrO_3 dissolved in water) plus a small, carefully controlled amount of **catalyst** — most commonly **sulfate** (SO_4), with the all-important ratio of CrO_3 : $\text{SO}_4 \approx 100$: 1. Some processes use a **mixed catalyst** (sulfate plus **fluoride**) for higher efficiency and speed.



TECHNICAL STUFF — THE 100:1 RATIO IS THE HEART OF THE BATH

Chromic acid alone won't deposit chrome; it needs a trace of catalyst to plate at all. But too much catalyst and the deposit goes dull, milky, or won't cover; too little and you get poor coverage and low efficiency. The classic target is CrO_3 : SO_4 of about **100:1 by weight** (e.g., ~250 g/L CrO_3 to ~2.5 g/L sulfate). Keeping that ratio in range is one of the most important bath-control jobs on the line — and it's done by your chemist/lab, not guessed at the tank.



FUN FACT

The catalyst is needed only in tiny amounts but is wildly powerful: a few grams per liter of sulfate makes the difference between a tank that plates beautiful hard chrome and one that won't plate at all. Chemistry at its most dramatic — a pinch of catalyst running a whole process.

• KEY TERMS TO CARRY FORWARD

- **Ampere-hour (amp-hour, A·h):** amps × hours of plating — the master measure of “how much” you've plated. Thickness is controlled by amp-hours (Part Three).
- **Throwing power:** the bath's ability to plate evenly into recesses and onto far surfaces. Hard chrome's is **poor** — fixed with conforming anodes and robbers (Part Three).
- **Conforming anode:** a custom anode shaped to match the part so current — and therefore chrome — lands evenly.
- **Robber / thief:** a piece of scrap metal placed to “steal” excess current away from edges so they don't over-build.
- **Microcracked deposit:** the fine, dense crack network in good hard chrome.



REMEMBER

The eight numbers that define this tank: CrO_3 ~225–300 g/L, **ratio 100:1**, **temp ~50–60°C**, **CD ~30–60+ A/dm²**, **efficiency ~12–25%**, **rate ~25 µm/hr**, **hardness 800–1100 HV**, **insoluble lead anodes**. Carry these and you can sanity-check almost anything about hard chrome.

CHAPTER 7

PPE — YOUR DAILY ARMOR

THE GEAR THAT STANDS BETWEEN YOU AND A CARCINOGENIC, CORROSIVE, HOT ACID

A hard chrome line works with hot caustic cleaners, strong acids, high-amperage current, and — above all — **carcinogenic Cr(VI) chromic acid and its mist**. PPE is your *last* layer of protection (after ventilation and mist suppression), but it is essential.

★

REMEMBER

There is no part, no order, and no deadline worth your health. Scrap can be reworked. Your lungs cannot. On this line that’s not a slogan — Cr(VI) makes it literal.

• THE MASTER PPE SET (WEAR AT ALL CHEMICAL STATIONS)

- **Chemical splash goggles** — sealed goggles, not safety glasses. Glasses leave gaps; chromic acid finds them.
- **Face shield** — over the goggles for any work at the chrome tank, etch, cleaner, or acid.
- **Chemical-resistant gloves** — elbow-length, rated for chromic acid and the cleaners (nitrile, neoprene, or PVC per the SDS). Inspect for pinholes before each use — a chrome ulcer starts at a single break.
- **Chemical-resistant apron/suit** — full coverage, chest to below the knee.
- **Chemical-resistant boots** — closed, non-slip, splash-proof.
- **Dedicated work clothing** — kept separate from street clothes; never worn home. This keeps Cr(VI) out of your car and your house.
- **Respiratory protection** — the respirator your shop’s Cr(VI) program specifies for the task. Required whenever you could be exposed to chromic acid mist above the limits, and any time ventilation is degraded.

⚠

HEADS-UP — RESPIRATOR IS A BACKUP, NOT A LICENSE

A respirator is required, but it does **not** replace tank ventilation and mist suppression. If the fume suppressant has died (foam gone, mist visible) or the exhaust isn’t pulling, the tank is unsafe even with a respirator on. Stop and report it. Engineering controls are the primary protection; the respirator is the last line.

💡

TIP

Put gear on *in order*: boots and apron first, then gloves, then goggles, then face shield and respirator last. Take it off in reverse, and rinse your gloves *before* removing them. This keeps contaminated gloves away from your face — critical with a carcinogen.

• STATION-BY-STATION HAZARD SUMMARY

STATION	PRIMARY HAZARD	WHY IT’S DANGEROUS
01 Inspect/Mask/Rack	Mechanical; solvent in some maskants	Cuts/pinches; solvent vapor
02 Clean/Degrease	Hot alkaline (caustic)	Chemical burns; hot splash
03 Rinse	Residual cleaner	Mild irritant; slip hazard
04 Anodic Etch	Strong acid / chromic etch; high current	Burns; possible Cr(VI); shock
05 Hard Chrome Plate	Cr(VI) chromic acid + mist; hot bath; very high amperage	Carcinogen; severe burns; respiratory harm; shock/arc-flash
06 Drag-Out/Rinse	Cr(VI)-contaminated water	Carcinogen; burns; slip hazard
07 Bake	Hot oven (~375°F)	Severe thermal burns
08 Grind/Finish	Chrome dust; rotating machinery	Inhalable Cr-bearing dust; entanglement/projectiles
09 Strip/Rework	Cr(VI); strong chemicals; current	Carcinogen; burns; shock
10 Dry/Inspect	Residual chemistry; hot air	Mild burns; slip

⚠

HEADS-UP

Never eat, drink, smoke, vape, or apply cosmetics on the line. Hand-to-mouth contact is a major route for ingesting Cr(VI). Wash thoroughly before breaks and before leaving. Keep all food and drink completely out of the work area, and keep work clothes separate from street clothes.

CHAPTER 8

EMERGENCY & FIRST-RESPONSE

WHEN SECONDS COUNT — LEARN THIS COLD, BEFORE YOU EVER NEED IT

Know these *before* you need them. **Locate, before your first shift:** the nearest **emergency eyewash** (within ~10 seconds of every chemical station), the **safety shower**, the **fire extinguisher** and alarm, the **rectifier emergency stop / disconnect**, the **spill kit**, the **SDS binder**, and your shop's **Cr(VI) regulated-area** boundaries and decon station.

**HEADS-UP**

Eyewash and shower stations must always be unblocked. Never stack parts, totes, or anode boxes in front of them. The day you need one is the day you can't move boxes out of the way.

• FIRST RESPONSE BY HAZARD

SITUATION	WHAT TO DO — IMMEDIATELY
Chromic acid / chemical on skin	Safety shower; flush at least 15 minutes . Remove contaminated clothing while flushing. Even a small chromic-acid contact on broken skin needs medical attention (chrome-ulcer risk). Bring the SDS.
Chromic acid / chemical in eyes	Eyewash; hold eyelids open and flush at least 15 minutes , rolling the eyes. Do not rub. Get medical attention immediately — chromic acid is corrosive to eyes. Bring the SDS.
Inhaled chromic acid mist, feel ill	Move to fresh air immediately. Report it — do not "tough it out." Seek medical attention for anything beyond brief mild irritation. Report it as a possible Cr(VI) exposure so monitoring/medical follow-up happens.
Cr(VI) swallowed	Do not induce vomiting unless the SDS directs it. Follow SDS first aid; call poison control / medical help immediately.
Nosebleed / sore nasal passages	Report it — recurrent nosebleeds or nasal soreness can be an early sign of Cr(VI) over-exposure. Don't ignore it; it triggers a ventilation/exposure check.
Electrical shock / arc-flash	Do not touch the person if they may still be in contact with current. Cut power at the emergency stop/disconnect first, then call for help and render aid.
Chemical spill	Alert others; keep people away. Clean only if trained and within your shop's "small spill" limits — otherwise evacuate and call trained spill response. Never let chromic-acid spills reach a floor drain — that's a reportable environmental release.

**HEADS-UP — LOCKOUT/TAGOUT (LOTO)**

Mandatory for *all* rectifier and electrical maintenance. Physically lock the power off and tag it so no one can re-energize while someone works. "I'll just be a second" is how people get electrocuted on a high-amperage chrome rectifier.

**HEADS-UP — A CRITICAL ACID-MIXING RULE, FOR LIFE**

When diluting acid (or dissolving chromic acid additions per your procedure), **always add ACID/chemical to WATER — never water to concentrated acid**. Adding water to concentrated acid releases heat so violently it can erupt acid into your face. The memory aid: "*Do as you oughta — add acid to water.*"

**FUN FACT**

The reason acid-to-water matters is heat capacity: a big pool of water soaks up the heat of dilution; a little water dropped onto acid flash-boils and spits. This one rule has saved more faces than any other in the chemical trades.

CHAPTER 9

THE SAFETY PHILOSOPHY OF A CR(VI) LINE

THE MINDSET THAT KEEPS YOU SAFE ACROSS A WHOLE CAREER

Rules and PPE are the *tools*. The philosophy is the *mindset* that makes them work. Six principles guide everything on this line — one more than the zinc manuals, because Cr(VI) earns it.

- 1. The hazard is invisible — so trust the controls, not your senses.** You can be over-exposed to chromic acid mist without seeing or smelling it. The ventilation, the foam blanket, and the air monitoring exist *because* your nose can't be trusted here. Believe the instruments.
- 2. Engineering controls come before PPE.** Suppress the mist at the source, ventilate the tank, *then* wear your respirator. PPE is the last layer, not the first. A respirator over a broken-ventilation tank is a false sense of safety.
- 3. Hygiene is a safety control, not housekeeping.** No eating, drinking, smoking, or face-touching; wash before breaks; keep work clothes separate. With a carcinogen, hand-to-mouth and take-home contamination are real exposure routes — your habits protect you and your family.
- 4. Containment beats cleanup.** Move parts deliberately; let them drain over the tank; don't fling racks or splash. Most exposures aren't dramatic spills — they're drips and aerosols from sloppy handling. Slow, smooth movement is safer *and* makes better parts (less drag-out, less mist).
- 5. Respect the two faces of every hazard.** The chromic acid that makes the hard, beautiful deposit is the same chromic acid that can give you cancer. The high current that plates it can arc-flash. You can't separate the work from the hazard — so you manage both at once, every time.
- 6. When in doubt, stop and ask.** The unsafe operator isn't the one who asks questions — it's the one who guesses. If the foam looks thin, a glove looks worn, the exhaust sounds weak, or a reading looks wrong, *stop and check*.

• THE HAZARD FAMILIES TO INTERNALIZE

- 1. Cr(VI) / chromic acid (the headline).** Carcinogenic mist, corrosive liquid, chrome ulcers, nasal damage. *Defense:* mist suppression, ventilation, respirator, full PPE, hygiene, OSHA 1910.1026 program.
- 2. Acids & alkali (corrosives).** Hot caustic cleaner and acid etch burn on contact and damage eyes. *Defense:* goggles, face shield, gloves, apron; 15-minute flush; acid-to-water.
- 3. Electrical & physical.** Very high-amperage rectifier (shock/arc-flash); wet/slick floors; grinding machinery; hot bake oven. *Defense:* LOTO, dry hands, non-slip boots, machine guards, heat awareness.



REMEMBER

Cr(VI) first, corrosives second, electrical/physical third. If you can name the hazard family at the tank in front of you — and on this line the first one is almost always Cr(VI) — you already know most of what you need to protect yourself.



REMEMBER

You now have the whole foundation: what plating is, what makes hard chrome a *functional* deposit, why the bath is hexavalent and why that's a carcinogen, how the line works as one system, the strange electrochemistry, what to wear, what to do in an emergency, and the safety mindset that ties it together. Every station chapter builds on this. Turn the page knowing the *why* — and especially the *danger* — and the *how* will make a lot more sense.

INSPECTION, MASKING & RACKING

“HARD CHROME GOES ONLY WHERE YOU PUT IT — AND THAT’S DECIDED BEFORE THE TANK.”

★ **REMEMBER — THE PREP PROMISE**

By the time a part reaches the chrome tank, most of whether it’ll turn out right has already been decided — at *this* station and at cleaning. As the old-timers say: “*The plating bath gets the credit, but the prep does the work.*”

• **PURPOSE**

Inspect the incoming part, **mask off every area that must NOT be plated**, and **rack it** so current reaches the right places evenly. In hard chrome this station is far more important than in zinc, because hard chrome has **terrible throwing power** — chrome piles onto edges and high spots and starves recesses — so *where the metal goes is controlled mechanically, here, before any chemistry.*

• **THE “WHY”**

Hard chrome is plated thick, expensive, hazardous, and usually ground to size. You don’t want it where it isn’t needed (it wastes costly chromic acid, adds grinding work, and can ruin tolerances on bearing journals, threads, and bores). **Masking** physically blocks chrome from those areas. **Racking** both carries current to the part and positions it relative to the anodes — and good racking is half of getting even thickness on a poor-throwing-power bath.

⚙️ **TECHNICAL STUFF — MASKING METHODS**

Common maskants: **plating tape** (PVC/lead tape) for edges and bands; **lacquers/stop-off paints** (peelable coatings) for irregular areas; **wax** for threads and blind holes; and custom **fixtures/plugs** for bores and faces. The maskant must survive a hot (50–60°C), strongly oxidizing chromic-acid bath without dissolving or lifting — ordinary tape won’t do.

★ **REMEMBER**

On hard chrome, **masking and rack position are your throwing-power tools**. You cannot fix poor coverage with chemistry alone the way a zinc operator can; you fix it here, with masks, conforming anodes, and robbers (Part Three).

• **MAKE-UP & KEY PARAMETERS**

ITEM	SPEC / PRACTICE	WHAT TO WATCH
Incoming inspection	Per traveler / customer print	Check for the hardness/high-strength flag (drives the bake decision)
Maskant	Chrome-rated tape, lacquer, wax, fixtures	Must withstand 50–60°C chromic acid; no lifting/dissolving
Surface prep before mask	Base must be sound; note pits, nicks, prior plating	Old chrome usually must be stripped first (Station 09)
Racking	Good electrical contact; firm, current-carrying	Loose contact = burning, no plating, dropped parts
Part orientation	Position relative to anodes for even current	Avoid trapping gas; allow drainage
Significant-surface ID	Mark surfaces that must hit thickness spec	These drive the amp-hour calc (Part Three)

• STEP-BY-STEP PROCEDURE

1. **Read the traveler first.** Note required chrome thickness, the **significant surfaces**, and any **high-strength / hardness flag** (it sets the mandatory bake at Station 07).
2. **Inspect the part.** Look for damage, deep pits, or **old chrome** that must be stripped before re-plating (route to Station 09 if so).
3. **Plan masking** — identify every surface that must stay bare of chrome (threads, bores, bearing faces, ends).
4. **Apply maskant** cleanly — tape edges down fully, paint lacquer evenly, plug/fixture bores. No gaps where chrome could creep, no loose edges that could lift in the hot bath.
5. **Rack the part** for solid electrical contact and correct orientation toward the anodes. Plan where **robbers** and **conforming anodes** will go (Part Three).
6. **Double-check** mask coverage and rack tightness. A part that drops in the chrome tank, or a mask that lifts, is a Cr(VI) handling problem as well as a scrap problem.
7. **Send to cleaning** promptly.



HEADS-UP

Some stop-off lacquers and tapes use **solvents** with their own vapor/flammability hazards — use them with ventilation and away from ignition sources, per their SDS. And handle racks carefully: pinch points, sharp edges, and heavy parts cause the everyday injuries here.

• TOP MISTAKES NEW OPERATORS MAKE

1. **Missing the hardness flag** on the traveler — leads to skipping the mandatory bake. This is a safety miss, not just a quality one.
2. **Sloppy masking** — gaps let chrome creep onto threads/bores; lifted edges let the hot bath under the mask.
3. **Using non-chrome-rated tape/lacquer** that dissolves in the hot chromic acid and contaminates the tank.
4. **Loose rack contact** — causes burning, missed plating, or a part dropping into the Cr(VI) tank.
5. **Ignoring throwing power** — not planning robbers/conforming anodes for edges and recesses.

• TEACH-BACK CHECKLIST

- ☐ Explain why masking matters more in hard chrome than in zinc (poor throwing power; thick, costly, ground-to-size deposit).
- ☐ Name three masking methods and one requirement they all share (survive hot chromic acid without lifting/dissolving).
- ☐ Explain what a “significant surface” is and why you identify it (it must hit thickness spec; drives the amp-hour calc).
- ☐ State what the hardness/high-strength flag triggers downstream (mandatory bake at Station 07).
- ☐ Explain why good rack contact matters (current, no burning, part doesn't drop into the Cr(VI) tank).

SIGN-OFF — STATION 01

Trainee: _____ Trainer: _____ Date: _____

“I confirm I checked the traveler (including the hardness flag), masked all non-plated surfaces with chrome-rated maskant, racked for solid contact and correct orientation, and identified the significant surfaces.”

STATION 02 OF 10

CLEAN / DEGREASE

“PLATE OVER DIRT AND YOU’VE JUST PLATED DIRT.”

• PURPOSE

Get the part **chemically clean** — no oil, grease, fingerprints, or shop soil — so the chrome bonds to bare steel.

• THE “WHY”

Chromium can only bond to clean, bare, active metal. Oil and shop soil block the bond; a thick hard-chrome layer that bonds to grease instead of steel will blister or peel — and a peeling 100 µm functional deposit is a real failure. Cleaning here usually means a **hot alkaline soak**, often followed by **alkaline electrocleaning** (running current to scrub the surface electrochemically) for stubborn soils. The free, foolproof check is the **water-break test**.



TIP — THE WATER-BREAK TEST (FREE, FAST, NEVER LIES)

Pull a cleaned part and watch the water. A truly clean surface sheets water in **one continuous, unbroken film** (PASS). A dirty surface makes water **bead up** (FAIL — re-clean). Costs nothing; takes two seconds; use it on every part.

FAIL - water beads up

```
+-----+
| o   o   o | X
|   o   o   |
+-----+
Oil remains -> RE-CLEAN
```

PASS - water sheets evenly

```
+-----+
|#####| OK
|#####|
+-----+
Surface clean -> proceed
```



TECHNICAL STUFF

“Water break” reflects the metal’s *surface energy*. Clean steel has high surface energy so water spreads (low contact angle); even a one-molecule-thick oil film drops the energy so water beads (high contact angle). That’s why it catches contamination far too thin to see.

• MAKE-UP & KEY PARAMETERS

PARAMETER	RANGE	WHAT TO WATCH
Type	Alkaline soak ± alkaline electroclean	Electroclean for heavy soils
Temperature	60–82°C (140–180°F)	Per cleaner TDS; don’t overheat
Time	3–10 min soak	Heavy oil longer; light oil shorter
Concentration	Per TDS (often 30–75 g/L)	Titrate regularly
pH	Strongly alkaline (≈11–13.5)	Burns on contact
Electroclean current	Per TDS (anodic/cathodic)	Polarity matters on hard steel (see Heads-Up)
Water-break test	Pass = continuous sheet	The only reliable field test



HEADS-UP — HOT ALKALINE SOLUTION

This tank is **hot AND caustic**. Splashes cause **chemical burns on contact**, and steam/mist irritates airways. Full PPE: splash goggles, face shield (when charging/dumping), elbow-length chemical gloves, apron, boots. Skin contact → flush 15 min; eyes → eyewash 15 min and get help.

**HEADS-UP — CATHODIC ELECTROCLEANING + HARD STEEL = HYDROGEN**

Cathodic electrocleaning puts hydrogen *into* the part (same embrittlement risk as the chrome tank). On high-strength steel, shops often use **anodic** (reverse) cleaning or a brief anodic finish to avoid charging the part with hydrogen before plating. Follow your shop's procedure for hard parts — this matters for embrittlement.

• STEP-BY-STEP PROCEDURE

1. **Check the tank:** temperature in range, concentration in range (last titration log), surface free of floating oil.
2. **Inspect the load** — heavy rust-preventive oil needs the long end of the time range.
3. **Immerse fully**, start the timer, confirm agitation.
4. **Electroclean if specified**, at the correct polarity for the part (anodic for hard steel where required).
5. **Withdraw slowly**, draining back into the tank.
6. **Run the water-break test.** Continuous sheet → next station. Beading → re-clean.
7. **Move promptly** so the clean surface doesn't sit and flash-rust.

• TOP MISTAKES

1. Skipping the water-break test because the part “looks fine” (the oil that ruins plating is invisible).
2. Running the tank too cold to cut oil.
3. Letting concentration drift low (cleaner gets used up).
4. Using **cathodic** electroclean on high-strength steel and loading it with hydrogen.
5. Letting clean parts sit and flash-rust before the etch.

• TEACH-BACK CHECKLIST

- ☐ State what the cleaner removes and what it does *not* (oil/soil — not oxide; that's the etch's job).
- ☐ Correctly perform and read a water-break test.
- ☐ State the temperature range and the caustic-burn hazard.
- ☐ Explain why cathodic electrocleaning can be a problem on hard steel (adds hydrogen).
- ☐ Name three pieces of required PPE.

SIGN-OFF — STATION 02

Trainee: _____ Trainer: _____ Date: _____

“I confirm the part passed the water-break test (continuous sheet), the cleaner was in range and at temperature, I used the correct electroclean polarity for the part's hardness, and I wore required PPE.”

STATION 03 OF 10

RINSE (THE FIREWALL)

“A RINSE ISN’T A REST STOP. IT’S A CHECKPOINT.”

PURPOSE

Wash the **alkaline cleaner film** off the part before it reaches the acidic etch/chrome chemistry.

THE “WHY”

The cleaner is alkaline; the etch and the chrome bath are acidic. Carry alkaline drag-out forward and it neutralizes and weakens the acid chemistry, wastes it, and degrades activation — showing up later as poor adhesion. The rinse knocks that drag-out down by dilution. We measure rinse cleanliness with **conductivity** ($\mu\text{S}/\text{cm}$) — *lower = cleaner*.



TIP — COUNTER-FLOW RINSING

The professional approach is a **counter-current** rinse: parts move forward, water flows backward. Fresh water enters the *last* (cleanest) tank and overflows into the dirtier one, so the cleanest water touches the part last. It saves enormous water — important on a line that already needs heavy water management for Cr(VI) control.

MAKE-UP & KEY PARAMETERS

PARAMETER	RANGE	NOTES
Temperature	Ambient	No heating needed
Time	30–60 s with agitation	Longer for bores/blind holes
Water source	Municipal or DI	DI preferred
Conductivity	< 500 $\mu\text{S}/\text{cm}$ (target < 300)	Monitor at shift start
Flow	Continuous overflow / counter-current	Prevents stagnation



HEADS-UP

The rinse carries dragged-in caustic and starts mildly alkaline. Wear goggles, gloves, non-slip footwear. **Wet floors and slips are the #1 everyday injury at rinses.**

STEP-BY-STEP PROCEDURE

1. Check conductivity at shift start; if near/above limit, tell your lead.
2. Confirm water is flowing.
3. Transfer the rack promptly from the cleaner (dried cleaner is harder to rinse).
4. Immerse fully with agitation, 30–60 s; longer for bores.
5. Lift and drain over the rinse tank.
6. Move promptly to the etch so the part doesn’t flash-rust.

TOP MISTAKES & TEACH-BACK

TOP MISTAKES

- Treating the rinse as a “quick dunk.”
- Ignoring the conductivity reading.
- Letting parts drip-dry between cleaner and rinse.
- Not flushing bores/blind holes.
- Skipping the drain step.

TEACH-BACK — CAN YOU ANSWER?

- Why carrying cleaner into the acid stages causes problems?
- Define drag-out / drag-in.
- What conductivity tells you and which way is “clean”?
- State the conductivity limit and target.

SIGN-OFF — STATION 03

Trainee: _____ Trainer: _____ Conductivity: _____ $\mu\text{S}/\text{cm}$

“I confirm the rinse conductivity was within limit, water flow was running, and parts were rinsed the full time with PPE worn.”

STATION 04 OF 10

ANODIC ETCH / ACTIVATION

“NO ACTIVATION, NO ADHESION. PERIOD.” — AND ON THIS LINE IT’S OFTEN DONE *IN THE CHROME TANK*

PURPOSE

Remove the invisible oxide film and a trace of base metal to leave a chemically fresh, micro-roughened steel surface for a **true metallurgical bond** with the chrome.

THE “WHY”

Even spotlessly clean steel grows an oxide film in minutes. Chrome can’t bond to oxide. In hard chrome, activation is usually an **anodic (reverse-current) etch**: the part is run as the **anode** (positive) so the surface is electrochemically etched/oxidized off, exposing fresh metal. This is frequently done **right in the chrome tank** — etch anodic for a short time, then *reverse polarity* to cathodic and plate — so the activated surface never sees air before chrome deposits on it. Some shops use a separate acid (e.g., sulfuric) reverse-etch tank instead.

**REMEMBER**

No activation, no adhesion — and on a thick, stressed hard-chrome deposit, lost adhesion can be a catastrophic failure, not just a cosmetic one. A 100 µm chrome layer that peels off a hydraulic rod is a serious problem. Activation has to come immediately before plating.

**TECHNICAL STUFF — ANODIC ETCH VS. CATHODIC ACTIVATION**

Running the part *anodic* etches metal *off* the surface (cleans + roughens, no hydrogen charged in). Running it *cathodic* would deposit/reduce and charge hydrogen in. The anodic-then-cathodic sequence in the chrome tank gives a fresh, well-bonded interface and is the classic hard-chrome activation. The etch current, time, and (if separate) acid are set by your process spec.

MAKE-UP & KEY PARAMETERS

PARAMETER	RANGE / PRACTICE	NOTES
Method	Anodic (reverse-current) etch	Often in the chrome tank ; or separate acid tank
Temperature	~50–60°C if in chrome tank	Matches plating bath
Etch current density	Per process spec (often high, anodic)	Set by your TDS/spec, not guessed
Time	Short — seconds to a couple minutes	Over-etch roughens; under-etch = no bond
Separate-tank option	e.g., H ₂ SO ₄ reverse etch	Avoids chrome exposure if not in-tank

**HEADS-UP**

If the etch is done **in the chrome tank**, you are already working over **carcinogenic Cr(VI) chromic acid** — full mist suppression, ventilation, respirator (per program), and PPE apply *now*, not just at “plating.” If it’s a separate **acid** tank, strong-acid burn and fume hazards apply, plus flammable hydrogen — no flames/sparks, and ventilation required.

**HEADS-UP — HIGH CURRENT + HOT ACID + REVERSE POLARITY**

Confirm the rectifier polarity is correct for etch vs. plate before energizing. Getting polarity backward wastes the load and can damage parts/anodes. LOTO before any rectifier work.

• STEP-BY-STEP PROCEDURE

1. Confirm whether activation is **in-tank** (chrome) or a **separate** acid tank, and set up accordingly.
2. Verify ventilation/mist suppression (if in chrome tank) and PPE/respirator.
3. Confirm rectifier set to **anodic (reverse)** for the etch step.
4. Immerse the part; run the anodic etch for the specified short time — *don't over-etch*.
5. **Switch polarity to cathodic** (if in-tank) and proceed straight into plating — no air exposure.
6. If separate tank: rinse minimally/transfer **fast** to the chrome tank before the surface re-oxidizes.

• TOP MISTAKES

1. Over-etching (roughens the surface; can pit the part).
2. Under-etching (oxide remains; the thick chrome peels later).
3. Wrong polarity (etch vs. plate) on the rectifier.
4. Letting an activated part sit and re-oxidize before plating.
5. Treating in-tank etch as “not really chrome yet” and skipping mist/respirator controls.

• TEACH-BACK CHECKLIST

- ☐ Explain what the etch removes and why (oxide + trace metal → fresh, bondable surface).
- ☐ Explain why it's anodic (etches off, doesn't charge hydrogen) and what happens next (reverse to cathodic, plate).
- ☐ Explain why it's often done in the chrome tank (no air exposure between activation and plating).
- ☐ State the Cr(VI) controls that apply during in-tank etch (mist suppression, ventilation, respirator, PPE).

SIGN-OFF — STATION 04

Trainee: _____ Trainer: _____ Etch: _____ s · In-tank? Y / N

“I confirm I activated the part by anodic etch for the specified time, with correct polarity, and (if in-tank) with full Cr(VI) controls in place, then moved straight into plating without air exposure.”



FUN FACT

The “etch then reverse and plate” trick in the same tank is one of plating's neatest moves: a single dip does two opposite electrochemical jobs — first stripping the surface clean as an anode, then growing chrome on it as a cathode — all without the part ever touching air in between.

STATION 05 OF 10 • THE MAIN EVENT

HARD CHROME PLATE

WHERE WORN STEEL BECOMES HARD, PRECISE, WEAR-RESISTANT STEEL — AND WHERE THE CARCINOGEN LIVES

Take a breath. Up to now, every station has had one job: get the part ready and protected. **Station 05 is where the hard chrome is actually created** — and it's the tank that defines this whole line, for performance *and* for danger.

The big idea in one sentence: **we use a lot of electricity to pull chromium out of hot chromic acid and build it, atom by atom, into a thick, hard, microcracked layer on your steel part.** The cast:

- **Cathode** — *your part* (negative). Where chrome lands.
- **Anode** — **insoluble lead / lead-tin (or lead-antimony) plates**. They carry current and keep the bath oxidized; they do **not** dissolve to feed chromium.
- **Electrolyte** — **chromic acid (CrO₃) + catalyst (sulfate, ~100:1)**, hot (~50–60°C), yellow, carcinogenic.

★

REMEMBER
Unlike zinc, **the anodes here do NOT feed metal**. All the chromium comes from the chromic acid you maintain as a chemical. The anodes are lead, insoluble, and just carry current. Forget this and the whole bath behaves like a mystery.

⚠

HEADS-UP — THIS IS THE SINGLE MOST DANGEROUS TANK ON THE ENTIRE LINE
Hot **carcinogenic Cr(VI)** chromic acid that throws an inhalable **mist**, very high amperage, corrosive liquid. **Mist suppression (foam/fume suppressant) + local exhaust (push-pull) must be working before you run it.** Respirator per program. Full PPE. No eating/drinking/face-touching. If the foam is gone or you can see mist rising, the tank is unsafe — stop and report.

• WHY HARD CHROME (AND WHY IT’S UNFORGIVING)

ADVANTAGE	WHAT IT MEANS ON THE FLOOR
Extreme hardness (800–1100 HV)	Survives heavy wear and abrasion
Low friction	Smoother sliding, less heat in moving parts
Build-up / salvage	Rebuild worn parts oversize, grind to spec
Corrosion + release properties	Resists many environments; molds release cleanly

...but the same chemistry brings hard problems: **very low efficiency (~12–25%)** (slow, current-hungry); **very poor throwing power** (needs conforming anodes/robbars/careful racking); **high hydrogen** → **embrittlement** (mandatory bake on hard steel); **high internal stress / microcracking**; and of course the **Cr(VI) hazard**.

• THE BATH MAKE-UP

KNOW THE RANGES

COMPONENT / PARAMETER	TYPICAL RANGE	WHAT IT DOES
Chromic acid (CrO ₃)	150–400 g/L (typ. 225–300)	The source of all the chromium
Catalyst — sulfate (SO ₄)	Set by ratio	Makes deposition possible
CrO ₃ : SO ₄ ratio	≈ 100 : 1	The heart of the bath — keep it in range
Optional mixed catalyst	Sulfate + fluoride	Higher efficiency/speed (some processes)
Temperature	~50–60°C (122–140°F)	Affects hardness, efficiency, crack pattern
Cathode current density	~30–60+ A/dm ² (300–600+ ASF)	Drives the (slow) deposition
Cathode efficiency	~12–25%	Most current makes gas, not chrome
Deposition rate	~25 μm/hr (typical)	Why thick jobs take hours
Anodes	Lead / lead-tin / lead-antimony (insol.)	Carry current; keep bath oxidized
Deposit hardness	800–1100 HV	The functional payoff
Mist suppressant	Fume suppressant / foam blanket	Cuts Cr(VI) mist at the source



TECHNICAL STUFF — TEMPERATURE AND CURRENT DENSITY WORK AS A PAIR

Hard chrome's appearance, hardness, and crack pattern depend on the *combination* of bath temperature and current density (often shown on a “plating range” chart for the process). Run too cold / too high CD and you can get dull or burnt deposits; too hot / too low CD and you can get milky, soft, or non-bright chrome with poor coverage. Your process spec defines the window — stay in it.



REMEMBER — THE NUMBERS TO CARRY

CrO₃ ~225–300 g/L (range 150–400), **ratio 100:1**, **temp ~50–60°C**, **CD ~30–60+ A/dm²**, **efficiency ~12–25%**, **rate ~25 μm/hr**, **hardness 800–1100 HV**, **insoluble lead anodes**. These eight facts let you sanity-check almost anything about this tank.

• THROWING POWER, CONFORMING ANODES & ROBBERS (PREVIEW)

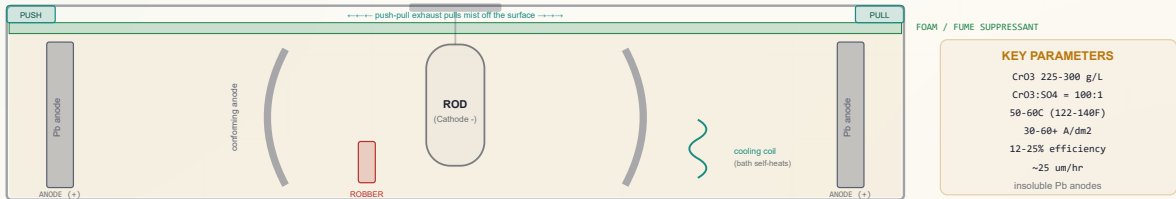
Hard chrome covers edges and high spots heavily and starves recesses and far surfaces. You manage it mechanically: **conforming anodes** (shaped to match the part so current lands evenly), **robbers/thieves** (scrap metal that “steals” current from edges so they don’t over-build), and **careful racking/orientation**. This is so central it gets its own Part Three section.



TECHNICAL STUFF

“Throwing power” is the bath’s ability to plate evenly into recesses and onto distant surfaces. Hard chrome’s is among the worst of any plating process — a direct consequence of its low efficiency and high stress. That’s why a hard-chrome rack often looks like a custom jig built around one specific part: the anodes and robbers *are* the throwing-power solution.

HARD CHROME TANK CROSS-SECTION — TANK ANATOMY



• INSOLUBLE ANODES — MAINTENANCE MATTERS

EQUIPMENT	SPEC	WHY IT'S THERE
Anodes	Lead / lead-tin / lead-antimony, insoluble	Carry current, keep bath oxidized ($\text{Cr}^{3+} \rightarrow \text{Cr}^{6+}$)
Anode film	Thin brown PbO_2 film (good)	Healthy anodes carry current evenly
Anode cleaning	Periodic per procedure	Yellow lead-chromate buildup raises resistance
Conforming anodes	Custom per part	Even current → even thickness
Cooling	Coil/heat exchanger	Bath self-heats from current; needs <i>cooling</i>
Ventilation	Push-pull / lateral exhaust + scrubber	Pull Cr(VI) mist off the surface
Mist suppressant	Foam blanket / surfactant	Suppress mist at the source



TECHNICAL STUFF — WHY INSOLUBLE ANODES NEED CARE

Lead anodes form a brown lead-dioxide (PbO_2) film that conducts and helps re-oxidize trivalent chromium back to hexavalent in the bath (keeping the chemistry healthy). But they can also grow an insulating *yellow* lead-chromate crust that raises resistance and causes uneven plating. Anodes that sit idle “go bad”; periodic cleaning/conditioning per your procedure keeps them working. Idle tanks are often kept “live” at low current to maintain the anode film.

• **STANDARD OPERATING PROCEDURE (STATION 05)**

1. **Confirm the part is clean, masked, racked, and freshly activated** (came straight from etch — ideally in-tank — with no air exposure).
2. **Verify Cr(VI) controls FIRST:** foam/fume suppressant present, push-pull exhaust running, scrubber on, your respirator and full PPE on.
3. **Check bath readiness:** temperature ~50–60°C (per spec), CrO₃ and catalyst ratio in range (per lab), anodes present/conditioned and not crusted, cooling working, conforming anodes/robbers placed.
4. **Confirm rectifier polarity is cathodic (plate)** and set current for the part's significant-surface area and target current density (~30–60+ A/dm²).
5. **Calculate the amp-hours** needed for the target thickness (Part Three) and set up the amp-hour counter / timer.
6. **(Often) anodic etch in-tank first**, then reverse to cathodic and begin plating.
7. **Plate for the calculated amp-hours** — typically *hours* for a functional build (~25 µm/hr). Monitor temperature, current, agitation, foam blanket, and exhaust the whole time.
8. **At target amp-hours, stop current, remove, drain over the tank**, and move to drag-out/rinse (Station 06).
9. **Log everything:** amp-hours, current, temperature, additions. The log is your thickness record and your bath-control history.

⚠

HEADS-UP — THE RECTIFIER IS VERY HIGH AMPERAGE

Hard chrome rectifiers push huge current. Shock and arc-flash risk. **LOTO before any maintenance, reaching into the tank, or work on bus bars.** Never with wet hands. No exceptions.

⚠

HEADS-UP — WATCH THE FOAM/EXHAUST THE WHOLE RUN

A hard-chrome plate runs for *hours*, and the foam blanket and exhaust must work the *entire* time, not just at the start. If the foam thins or mist becomes visible mid-run, you are being exposed to a carcinogen. Re-dose suppressant / fix exhaust per procedure, or stop. Long runs are exactly when complacency creeps in — don't let it.

• **COMMON DEFECTS — WHAT YOU'LL SEE AND WHAT IT MEANS**

WHAT YOU SEE	MOST LIKELY CAUSE	WHAT TO DO
Dull / milky deposit	Temp/CD off the window; ratio off; low CrO ₃	Check temp, CD, and CrO ₃ :SO ₃ ratio (lab)
Burnt, rough at edges/tips	CD too high there; no robbers	Add robbers; adjust racking/CD
Bare/skip plating in recesses	Poor throwing power (normal); needs conforming anode	Add conforming anode; re-rack
Poor adhesion / peeling	Bad activation; surface not bare/active	Re-etch; check cleaning; verify in-tank activation
Blistering / skip plating	Cr ³⁺ too high, or contamination	Lab check; oxidize/treat per chemist
Coarse macrocracks (not fine micro)	Temp/CD/ratio drift; high stress	Lab review of bath parameters
Slow build / low rate	Low CD; low efficiency; cool bath	Verify CD, temperature, ratio
No plating at all	Wrong polarity; crusted anodes; ratio way off	Check polarity; condition anodes; lab check

INTEGRATED SAFETY — THIS STATION DEMANDS TOTAL RESPECT



HEADS-UP — CR(VI) CHROMIC ACID + MIST (THE HEADLINE)

Carcinogenic, corrosive, hot. Engineering controls first (foam blanket + push-pull + scrubber), respirator second, full PPE always. Any skin/eye contact → flush 15 min and get medical attention (chrome-ulcer / eye-corrosion risk). Report any nosebleed or nasal soreness.



HEADS-UP — HIGH-AMPERAGE RECTIFIER

Shock/arc-flash. LOTO for all electrical/maintenance work. Dry hands only.



HEADS-UP — HYDROGEN EMBRITTLEMENT

All that hydrogen at the part soaks into hard steel. **High-strength parts MUST be baked (Station 07) within hours.** Confirm the bake routing now — flag it on the traveler.

TEACH-BACK & TOP MISTAKES — STATION 05

TEACH-BACK — CAN YOU ANSWER?

- Which electrode is your part, and which charge? (Cathode, -.)
- Why don't the anodes feed metal here? (Insoluble lead.)
- The headline bath numbers (CrO₃, ratio, temp, CD, eff., rate, hardness)?
- Why is efficiency so low — and why does that mean lots of hydrogen?
- What fixes poor throwing power? (Conforming anodes, robbers, racking.)
- Why must the foam/exhaust work the *whole* run?
- Mandatory downstream step for high-strength steel? (Bake.)
- The five Cr(VI) controls?

TOP MISTAKES TO AVOID

- Running with a thin/absent foam blanket or weak exhaust.
- Treating it like a quick dip (it's an hours-long, low-eff. run).
- Ignoring the CrO₃:SO₄ ratio (the heart of the bath).
- Using crusted/idle-damaged lead anodes.
- Forgetting robbers/conforming anodes (burnt edges/starved recesses).
- Skipping or delaying the bake on hard steel.
- Eating/drinking/face-touching near the tank.

SIGN-OFF — STATION 05

"I confirm I can independently: verify Cr(VI) engineering controls (foam suppressant, push-pull exhaust, scrubber) and PPE/respirator before running; verify bath readiness (temp, CrO₃:SO₄ ratio per lab, anode condition, cooling, conforming anodes/robbers); set correct polarity and current density; control thickness by ampere-hours; recognize common defects; and state the mandatory bake requirement for high-strength steel. I understand chemistry adjustments are made only by authorized personnel."

Operator: _____ Date: _____ Trainer: _____ Date: _____

STATION 06 OF 10

DRAG-OUT & RINSE

“RECOVER THE CHROME, CONTAIN THE CARCINOGEN, WASH THE PART.”

PURPOSE

Remove the chromic-acid film clinging to the freshly plated part — and, where possible, **recover** that dragged-out chrome so it can return to the tank instead of becoming hazardous waste.

THE “WHY”

On a Cr(VI) line, drag-out is doubly serious: it wastes costly chromic acid *and* it spreads a carcinogen into rinse water and, ultimately, the wastewater stream. So hard chrome lines use a **drag-out recovery rinse** (sometimes a still “dead” rinse, or a counter-current cascade) right after plating: the first rinse is kept fairly concentrated and is periodically returned to the chrome tank (helped by the fact the tank self-evaporates because it runs hot). Cleaner rinses follow.



REMEMBER

A post-chrome rinse isn’t “just a rinse” — it’s **chrome recovery and carcinogen containment**. The closer you keep that first recovery rinse to the tank, the more chrome you save and the less Cr(VI) escapes into the waste system.



TECHNICAL STUFF — WHY RECOVERY WORKS ON CHROME

Because the chrome tank runs hot, it constantly *evaporates* water. That evaporation makes room to pour the concentrated first-rinse (drag-out) water back into the tank, recovering both the chemical and the water. It’s a neat closed-loop trick that’s far more practical on a hot chrome tank than on a room-temperature zinc tank.

MAKE-UP & KEY PARAMETERS

PARAMETER	RANGE / PRACTICE	NOTES
Recovery (drag-out) rinse	Still/dead or first cascade stage	Returned to chrome tank periodically
Following rinses	Counter-current cascade	Drive conductivity down
Conductivity (final rinse)	Low (per shop spec)	Confirms chrome washed off
Temperature	Ambient	—
Drain time over tank	A few seconds	Cuts drag-out sharply



HEADS-UP — THIS RINSE WATER IS CR(VI)-CONTAMINATED

The recovery and first rinses contain hexavalent chromium. Treat them as hazardous: full PPE, no splashing, **never to a floor drain**. All Cr(VI)-bearing rinse water goes to the chrome **waste-treatment** system (Part Three), never to the sewer untreated.

STEP-BY-STEP & TOP MISTAKES

STEP-BY-STEP

- Drain the part over the chrome tank a few seconds (return chrome).
- Dip in the recovery (drag-out) rinse (captures concentrated chrome).
- Move through the following cascade rinses to wash clean.
- Confirm well rinsed (per shop check) before the bake oven.
- Return concentrated recovery rinse to the tank as it evaporates.

TOP MISTAKES

- Not draining over the tank first (wastes chrome, spreads Cr(VI)).
- Letting Cr(VI) rinse water reach a drain (a violation).
- Treating recovery water as plain rinse water.
- Under-rinsing before the bake (chromic acid bakes onto the part).

SIGN-OFF — STATION 06

Trainee: _____ Trainer: _____ Date: _____

“I confirm I drained over the tank, used the recovery rinse, washed the part through the cascade, kept all Cr(VI) water out of drains, and returned recovery water per procedure.”

STATION 07 OF 10

HYDROGEN-EMBRITTLEMENT BAKE

“DRIVE THE HYDROGEN OUT — BEFORE IT CRACKS THE PART.”

PURPOSE

Bake plated high-strength/hardened steel parts to drive absorbed hydrogen back out of the steel, preventing delayed brittle cracking in service.

THE “WHY”

Hard chrome’s low efficiency makes a *lot* of hydrogen at the part, and on high-strength steel some diffuses into the lattice and embrittles it — so a part can **crack and fail suddenly under load, hours or days later**. A controlled bake gives the hydrogen energy to diffuse back out *before* it does damage. This is a **mandatory** step for high-strength steel, not a judgment call.

**REMEMBER**

On high-strength steel, the bake is not optional. Skipping it can turn a perfect-looking part into a hidden failure. The hardness flag from Station 01 is what routes a part to this oven — never let a flagged part skip the bake.

**TECHNICAL STUFF — TYPICAL BAKE**

A common embrittlement-relief bake is **~190°C (375°F) for at least 8 hours**, started **soon after plating** (commonly within a few hours — your spec, often referencing **ASTM B850**, sets exact temp, time, and the maximum delay). Higher-strength steels need longer bakes; wait too long and hydrogen can crack the part *before* you ever bake it.

MAKE-UP & KEY PARAMETERS

PARAMETER	TYPICAL	NOTES
Temperature	~190°C (375°F)	Per spec / ASTM B850; varies with strength
Time	≥ 8 hours (often longer)	Longer = more hydrogen removed
Start delay after plating	Short – within hours	Don’t let plated hard parts sit
Trigger	High-strength / hardened steel flag	From the traveler (Station 01)
Logging	Time/temp record per part/lot	Quality + traceability

**HEADS-UP — HOT OVEN**

The bake oven runs hot enough to cause **severe thermal burns**. Use heat-resistant gloves, watch for hot racks/fixtures, and follow oven procedures for loading/unloading.

**HEADS-UP — DO NOT DELAY THE BAKE**

A high-strength part left overnight before baking may already have cracked. Plate, rinse, and **bake promptly** — treat the time window as a hard requirement on flagged parts.

STEP-BY-STEP, TOP MISTAKES & TEACH-BACK

STEP-BY-STEP & TEACH-BACK

- Confirm hardness flag & required bake (temp/time/delay).
- Get the part in the oven promptly after plating/rinse.
- Set oven ~375°F; load without crowding (even heat).
- Bake the full time (≥ 8 hr); log time & temp; cool; route to grind.
- Can you say why low efficiency → embrittlement?*

TOP MISTAKES

- Skipping the bake on a flagged high-strength part.
- Delaying the bake too long after plating.
- Baking too short or too cool to remove the hydrogen.
- Crowding the oven so parts heat unevenly.
- Not logging the bake (no proof it was done to spec).

SIGN-OFF — STATION 07

Trainee: _____ Trainer: _____ Bake: _____ °F / _____ hr · Delay: _____

“I confirm I verified the hardness flag, started the bake within the allowed delay, baked to spec temperature and time, and logged it.”

STATION 08 OF 10

GRIND / FINISH TO SIZE

“HARD CHROME IS PLATED OVERSIZE ON PURPOSE — GRINDING BRINGS IT TO SPEC.”

PURPOSE

Machine (usually **grind**) the deliberately oversize chrome deposit down to the **final, precise dimension and surface finish** the customer requires.

THE “WHY”

Because hard chrome is a *functional* deposit on precision parts, it’s plated slightly **oversize** and then ground to the exact diameter/dimension and surface finish — this is the step that makes hard chrome “dimensional.” Chrome is very hard (800–1100 HV), so it’s ground with appropriate wheels, light passes, and plenty of coolant to avoid heat damage and grinding cracks.

★

REMEMBER
Plate oversize, grind to size. This is the defining habit of functional hard chrome and the reason masking and even thickness (throwing power) matter so much: if a recess plated thin, grinding the high spots to size can leave the low spot *undersize* and scrap.

⚙️

TECHNICAL STUFF — GRIND GENTLY
Aggressive grinding overheats the chrome and can cause **grinding cracks** or burns in the deposit. Use the specified wheel, light depth of cut, adequate coolant, and don’t dwell. On critical parts a final polish or superfinish may follow. Grinding parameters are set by your shop’s machining spec.

MAKE-UP & KEY PARAMETERS

PARAMETER	PRACTICE	NOTES
Process	Cylindrical/surface grinding (then polish if specified)	Chrome is very hard
Wheel / coolant	Per machining spec; ample coolant	Prevent heat/grinding cracks
Stock removal	Light passes to final size	Don’t grind through to base
Final dimension	Per print tolerance	Check with proper gauging
Surface finish	Per print (Ra)	Polish/superfinish if required

⚠️

HEADS-UP — CHROME DUST AND ROTATING MACHINERY
Grinding generates **chrome-bearing dust** (don’t breathe it — use dust collection/PPE per your shop) and the usual rotating-machinery hazards: entanglement, projectiles, pinch points. Use guards, eye protection, no loose clothing/jewelry, and follow machine LOTO for setup/changeover.

STEP-BY-STEP, TOP MISTAKES & TEACH-BACK

STEP-BY-STEP & TEACH-BACK

- Confirm the part was baked (if flagged) *before* grinding.
- Set up the grinder per spec (wheel, coolant, work-holding).
- Take light passes, measuring frequently to final size.
- Don’t grind through chrome into base metal.
- Gauge to final tolerance & finish; route to inspection.
- Why plate oversize / grind to size? Why gently?*

TOP MISTAKES

- Grinding before the (required) bake.
- Aggressive grinding causing grinding cracks/burn.
- Grinding through to base metal (undersize/scrap).
- Not accounting for uneven plating (low spot undersize).
- Skipping dust controls (breathing chrome dust).

SIGN-OFF — STATION 08
Trainee: _____ Trainer: _____ Final dim: _____ · Finish: _____
“I confirm the part was baked (if required) before grinding, I ground to final dimension and finish without burning/cracking, used dust and machine-guarding controls, and gauged to print.”

STATION 09 OF 10

STRIP & REWORK PATH

“BAD CHROME COMES OFF, THEN GOES BACK ON — DONE SAFELY.”

• PURPOSE

Remove existing (worn, damaged, or out-of-spec) chrome so a part can be **re-plated** — the salvage/rework path that makes hard chrome economical.

• THE “WHY”

Hard chrome’s whole value proposition is rebuilding parts. When old or rejected chrome must come off before re-plating, it’s **stripped** — commonly by **reverse-current (anodic) stripping** in an alkaline or chromic solution (running the part anodic so the chrome dissolves off), or by a chemical strip per your process. After stripping, the part re-enters the line at cleaning/activation.



REMEMBER

Stripping is **reverse plating** — run the part *anodic* and the chrome comes off. It’s the same electrochemistry you use to plate, just pointed the other way.



TECHNICAL STUFF

Reverse-current stripping in an alkaline solution avoids attacking the steel base while removing chrome, and (when it produces Cr(VI)) feeds the same waste-treatment stream. Some shops strip in a dedicated tank; others use the chrome tank anodically with care. Either way it’s a Cr(VI)-relevant operation: the chromium going into solution is hexavalent and must be treated.

• MAKE-UP & KEY PARAMETERS

PARAMETER	PRACTICE	NOTES
Method	Reverse-current (anodic) strip; or chemical strip	Per process spec
Solution	Alkaline or chromic, per procedure	Choose to spare base metal
Polarity	Anodic (part positive)	Opposite of plating
After strip	Inspect base; re-clean/activate	Re-enter line at Station 02/04
Waste	Cr(VI)-bearing → treatment	Never to drain



HEADS-UP — CR(VI) AND STRONG CHEMICALS + CURRENT

Stripping involves carcinogenic chromium, strong solutions, and high current — all the chrome-tank hazards apply: ventilation, mist suppression where relevant, respirator/PPE, LOTO for rectifier work, and Cr(VI) waste handling.

• STEP-BY-STEP, TOP MISTAKES & TEACH-BACK

STEP-BY-STEP & TEACH-BACK

- Confirm the part needs stripping and the correct method.
- Set up: correct solution, **anodic** polarity, controls/PPE.
- Strip until chrome is removed, watching the base metal.
- Inspect the bare base for damage/pitting.
- Re-enter at clean/activate, then re-plate.
- Route strip solution/rinses to Cr(VI) treatment.

TOP MISTAKES

- Wrong polarity (won’t strip / damages part).
- Over-stripping into the base metal.
- Treating strip waste as non-hazardous.
- Re-plating without re-cleaning/re-activating.

SIGN-OFF — STATION 09

Trainee: _____ Trainer: _____ Date: _____

“I confirm I stripped the chrome at correct (anodic) polarity without damaging the base, applied Cr(VI) controls and PPE, routed waste to treatment, and re-cleaned/activated before re-plating.”

STATION 10 OF 10

DRY / FINAL INSPECTION

“SEND IT OUT RIGHT — DRIED, MEASURED, AND PROVEN TO SPEC.”

PURPOSE

Dry the finished part and **verify** it meets spec — thickness, hardness, adhesion, surface finish, and crack pattern — before it ships.

THE “WHY”

A functional hard-chrome part is a precision, high-value item; the customer is paying for a deposit that’s the right thickness, hard enough, well-bonded, and properly finished. Final inspection is where you *prove* that, and catch anything that needs rework before it leaves.

MAKE-UP & KEY QUALITY CHECKS

CHECK	METHOD (TYPICAL)	LOOKING FOR
Thickness	Micrometer (before/after grind), magnetic/X-ray gauge, cross-section	Meets print on significant surfaces
Hardness	Microhardness (Vickers/Knoop) on a coupon/cross-section	800–1100 HV range
Adhesion	Bend/file/thermal-shock test on a sample	No flaking/peeling
Crack pattern	Microscope on a sample	Fine microcracks (not coarse macro)
Surface finish	Profilometer / visual	Meets Ra spec
Dimension	Print gauging	Within tolerance

**HEADS-UP**

Even at “the end,” parts may carry residual chemistry and the dryer is hot — wear PPE, watch for hot-air/surface burns, and keep handling clean (a finished chrome part should not be re-contaminated).

STEP-BY-STEP, TOP MISTAKES & TEACH-BACK

STEP-BY-STEP & TEACH-BACK

- Dry the part gently and completely.
- Gauge the dimension to final tolerance.
- Verify thickness on significant surfaces.
- Check hardness, adhesion, crack pattern (often on coupons).
- Inspect surface finish/appearance.
- Accept & ship, or route to rework. Log results.

TOP MISTAKES

- Shipping without verifying thickness on the significant surface.
- Confusing normal microcracking with a defect (or missing macrocracks).
- Re-contaminating a finished part by careless handling.
- Not logging inspection results (no traceability).

SIGN-OFF — STATION 10

Trainee: _____ Trainer: _____ Date: _____

“I confirm I dried the part, verified thickness/dimension/hardness/adhesion/finish per the inspection plan, distinguished normal microcracking from defects, logged results, and routed any out-of-spec parts to rework.”

PART THREE — ACROSS THE LINE
THE CORE OPERATOR MATH

THICKNESS CONTROL BY AMPERE-HOURS

“AMP-HOURS MAKE MICRONS.”

Hard chrome thickness isn't controlled by “plate until it looks right” — it's controlled by **ampere-hours (A·h)**: amps multiplied by hours of plating. Because the deposit is thick and the process is slow and low-efficiency, getting thickness right means *counting amp-hours*, not watching the clock alone.

THE CHAIN OF LOGIC (FARADAY + EFFICIENCY)

1. **Faraday's law** says the *amount of metal deposited is proportional to the electrical charge passed* (amp-hours).
2. But hard chrome is only **~12–25% efficient**, so only that fraction of the current actually makes chrome — the rest makes gas.
3. So: **thickness depends on amp-hours × area × efficiency**. For a given part area and a known bath efficiency, a fixed number of **amp-hours per unit area** produces a known thickness.



REMEMBER

Amp-hours make microns. Set the current for your target current density, calculate the amp-hours needed for the required thickness on the significant surface, run that many amp-hours, and you hit thickness. This is *the* defining operator skill of hard chrome.



TECHNICAL STUFF — THE PRACTICAL VERSION

In practice you don't recompute Faraday's law each time; your shop has a **chart or factor** (“X amp-hours per dm² (or ft²) per micron (or mil) at this bath's efficiency”). You: (1) find the part's plated **area**, (2) multiply by the **target thickness factor** to get total amp-hours, (3) set the **current** (area × target current density), (4) run until the **amp-hour counter** reaches your number. Because efficiency drifts a little with temperature/CD/age, shops verify with a thickness measurement and adjust the factor.



TIP

An **amp-hour meter** (integrating ampere-hour counter) on the rectifier is your best friend — it tracks charge passed automatically. If you only have amps and a clock, amp-hours = average amps × hours, but real loads vary, so a counter is far more accurate for the long runs hard chrome needs.



FUN FACT

Faraday worked out the law linking electricity and deposited metal in the 1830s — long before anyone hard-chromed a hydraulic rod. Every time you set amp-hours for a thickness, you're using 190-year-old physics that still runs the shop floor.



HEADS-UP

Low efficiency cuts both ways: it means **long plating times** *and* that small efficiency changes shift your real thickness. Always **verify thickness** on the significant surface (Station 10) rather than trusting the amp-hour math blindly on a critical part.

— BEATING HARD CHROME'S WORST WEAKNESS

CONFORMING ANODES, ROBBERS & THROWING POWER

HOW YOU BEAT UNEVEN COVERAGE — WITHOUT CHEMISTRY

Hard chrome has **very poor throwing power**: current (and therefore chrome) piles onto edges, corners, and high spots and starves recesses, bores, and far surfaces. You can't fix this with additives the way a zinc operator can — you fix it **mechanically**, with three tools:

- **Conforming anodes** — custom anodes **shaped to follow the part's contour** so the gap (and thus current) between anode and part is even everywhere. A conforming anode for a rod is a tube/curved plate around it; for a bore, an internal anode goes *inside* the hole so the inside actually plates.
- **Robbers (thieves)** — pieces of scrap metal placed near **edges and corners** to “steal” excess current away from those high-current spots so they don't over-build (burn) — making the *rest* of the part more even.
- **Racking / orientation** — positioning the part relative to the anodes, spacing, and using auxiliary anodes for recesses; orienting to avoid trapped gas.



REMEMBER

Conforming anodes even out coverage; robbers steal current from edges so they don't burn; good racking ties it together. On hard chrome, the anode setup is custom-built around the part — that's normal, and it's *the* answer to poor throwing power.



TECHNICAL STUFF — BORES NEED INTERNAL ANODES

Because throwing power is so poor, a deep bore or ID will get **no chrome at all** from outside anodes — current can't reach in. To plate an inside diameter you must put an **internal (conforming) anode** down the bore, often with its own flow/agitation. Many “this won't plate inside” problems are simply a missing internal anode.



TIP

If a part keeps coming out **thin/bare in recesses**, think *throwing power first*: add a conforming/auxiliary anode for that area, not more current (more current just burns the edges worse). If **edges keep burning**, add a *robber*. The two problems usually travel together.



HEADS-UP

Robbers and auxiliary anodes are extra metal in a high-current Cr(VI) tank — handle, place, and remove them carefully (current, hot acid, carcinogen). LOTO/de-energize before reaching in to adjust them.

DEEP DIVE

HYDROGEN EMBRITTLEMENT & BAKING

THE MOST IMPORTANT SAFETY-AND-QUALITY CONCEPT ON THE LINE, IN FULL

What it is: During plating (and acid/etch steps), atomic hydrogen generated at the part diffuses into the steel and wedges between iron atoms. On **high-strength/hardened steel** this makes the metal **brittle**, so it can **crack and fail suddenly under load — hours or days later**, with no warning. On a critical fastener, rod, or structural part, that's a real-people safety failure.

Why hard chrome is high-risk: its **low efficiency (~12–25%)** means a *lot* of hydrogen is generated at the cathode per unit of chrome — more chance for hydrogen to soak in than in a high-efficiency bath.

Who's at risk: generally **high-strength steel, roughly ≥ 40 HRC / ≥ 180 ksi tensile**. The traveler's hardness flag (Station 01) identifies these.

THE FIX — BAKING (DE-EMBRITTLEMENT / HYDROGEN RELIEF)

- Typically **$\sim 190^{\circ}\text{C}$ (375°F) for ≥ 8 hours**, started **promptly after plating** (within a few hours), per **ASTM B850** / customer spec.
- Higher-strength steels need **longer** bakes.
- The **delay matters** — wait too long and the part can crack *before* baking.



REMEMBER

Low efficiency → lots of hydrogen → mandatory bake on high-strength steel. The chain is the whole story. The hardness flag routes the part; the bake ($\sim 375^{\circ}\text{F}$, ≥ 8 hr, promptly) saves it.



TECHNICAL STUFF — TERMS

HRC = Rockwell C hardness (higher = harder). **ksi UTS** = thousands of psi of ultimate tensile strength. **ASTM B850** is the standard guide for post-coating hydrogen-embrittlement relief baking — it sets temperature, time, and timing by strength level. You don't calculate these — you recognize the flag and follow the routing.



HEADS-UP

Embrittlement is a *delayed, hidden* failure — the part looks perfect, passes inspection, then fails in service. That's exactly why the bake is **mandatory and not skippable** on flagged parts, and why you don't let plated hard parts sit before baking.

ACROSS THE LINE

GRINDING, RINSING & WATER MANAGEMENT

TURNING AN OVERSIZE DEPOSIT INTO A PRECISE DIMENSION — AND KEEPING WATER UNDER CONTROL

• GRINDING & FINISHING TO SIZE (BASICS)

Functional hard chrome is plated **oversize** and **ground to final size** — that's what makes it dimensional. Basics every operator should know:

- **Grind gently:** chrome is very hard; aggressive grinding overheats it and causes **grinding cracks/burn**. Use the specified wheel, **light passes**, ample **coolant**, no dwelling.
- **Watch remaining thickness:** don't grind **through** the chrome into the base metal (undersize = scrap).
- **Uneven plating bites here:** if a recess plated thin (throwing power), grinding the high spots to size can leave the low spot **undersize**. Even plating up front prevents grind-stage scrap.
- **Bake before grind:** grind the part in its final, hydrogen-relieved, stress-settled condition.
- **Finish:** a polish/superfinish may follow to hit the surface-finish (Ra) spec.



REMEMBER

Plate **oversize**, **bake**, then **grind to size**. Order matters: bake before final grinding so the part is metallurgically settled at its final dimension.



HEADS-UP

Grinding chrome makes **chrome-bearing dust** — don't breathe it (dust collection/PPE), and respect rotating-machinery hazards (guards, eye protection, no loose clothing, machine LOTO for setup).

• RINSING & WATER MANAGEMENT

On a hard-chrome line rinsing does triple duty: it **protects the next chemistry**, **recovers dragged-out chromic acid** (saving money), and **contains Cr(VI)** so it doesn't spread or reach the wastewater untreated.

- **Counter-current (cascade) rinsing** saves water — fresh water enters the cleanest stage and flows backward toward the dirtiest.
- **Drag-out recovery rinses** after plating capture concentrated chrome for return to the (hot, evaporating) tank.
- **Conductivity** is the cleanliness gauge — *lower = cleaner*.
- All **Cr(VI)-bearing rinse water** goes to **waste treatment**, never to a drain.



REMEMBER

Every rinse is a **checkpoint**, not a rest stop. After the chrome tank, the first rinse is **chrome recovery + carcinogen containment** before it's anything else.



HEADS-UP

Never let chromic-acid or Cr(VI) rinse water reach a floor drain — that's a reportable environmental release. Keep drains protected and route all chrome-bearing water to treatment.

— RENDERING THE CARCINOGEN SAFE

CR(VI) WASTE TREATMENT & AIR PERMITS

WHERE THE CARCINOGEN IS RENDERED SAFE — CHEMISTRY EVERY OPERATOR SHOULD UNDERSTAND

A hard-chrome shop can't simply pour spent chemistry or rinse water down the drain. Cr(VI) must be **treated** to make it safe, the air must be **controlled and permitted**, and the waste must be **handled as hazardous**.

• THE CORE REACTION: REDUCE, THEN PRECIPITATE

The standard wastewater treatment for hexavalent chromium is a **two-step**:

1. **Reduce $\text{Cr}^{6+} \rightarrow \text{Cr}^{3+}$.** Add a **reducing agent** (commonly sodium metabisulfite, sulfur dioxide, or ferrous sulfate) at **low pH (acidic)**. This converts the dangerous, soluble hexavalent chromium into the far-less-hazardous **trivalent** chromium. (You can often see the color shift from yellow toward green.)
2. **Precipitate as hydroxide and remove. Raise the pH** (add caustic/lime) so the trivalent chromium drops out of solution as insoluble **chromium hydroxide sludge**, which is then **settled/filtered** out. The treated water is tested before discharge; the chromium sludge is managed as **hazardous waste**.



REMEMBER

Reduce then precipitate: $\text{Cr}^{6+} \rightarrow \text{Cr}^{3+}$ (acidic + reducing agent), then raise pH to drop out the chromium as sludge and filter it. This is the single most important environmental concept on a hard-chrome line.



TECHNICAL STUFF

The reduction needs **acidic** conditions to go fast and complete; the precipitation needs **alkaline** conditions. So the treatment swings pH down (reduce) then up (precipitate) — and the final discharge pH and chromium level must meet your permit limits before any water leaves. The trivalent-hydroxide sludge is a regulated **hazardous waste** with manifesting and disposal requirements.

• AIR PERMITS & MIST CONTROL

Because chromic-acid mist is a regulated carcinogen, hard-chrome tanks are controlled at the source (**fume suppressants/foam, push-pull ventilation**) and the exhausted air is usually cleaned by a **fume scrubber** before release. The operation typically requires an **air permit**, and there are federal emission rules specifically for chromium electroplating. Your shop's permit dictates the controls — keeping the foam blanket and exhaust working is part of staying compliant *and* keeping you safe.



HEADS-UP

These controls protect **you** (the mist you don't breathe) *and* the **public/environment** (the chromium that doesn't leave the stack or the drain). A dead foam blanket or a bypassed scrubber is both a personal exposure and a permit violation. Treat them as non-negotiable.



FUN FACT

The yellow-to-green color change during chromium reduction is the same chemistry your eyes can track in the waste treatment tank: yellow = hexavalent (dangerous), green = trivalent (much safer). The treatment plant is literally turning the carcinogen into something far less harmful, one color shift at a time.

— PROVING THE DEPOSIT IS RIGHT

QUALITY CHECKS

THICKNESS, HARDNESS, ADHESION, CRACK PATTERN

CHECK	WHAT "GOOD" LOOKS LIKE	HOW IT'S CHECKED
Thickness	Meets print on significant surfaces; even enough to grind to size	Micrometer (pre/post grind), magnetic/X-ray gauge, cross-section
Hardness	800–1100 HV	Microhardness (Vickers/Knoop) on coupon/cross-section
Adhesion	No flaking/peeling; bonded to steel	Bend, file, or thermal-shock test on a sample
Crack pattern	Fine, dense microcracks	Microscope on a sample — <i>macrocracks are the defect</i>
Surface finish	Meets Ra spec after grind/polish	Profilometer / visual
Dimension	Within print tolerance	Gauging



REMEMBER

In hard chrome, a **microcracked** deposit is a *pass*, not a fail. Watch for the wrong pattern — **coarse macrocracks**, or too few cracks — which signal bath/parameter drift.



TECHNICAL STUFF

Many of these checks run on **test coupons** plated alongside the lot (so you don't destroy the real part). The coupon gets cross-sectioned for thickness/hardness/crack count and bent for adhesion. Significant-surface thickness on the actual part is verified non-destructively.

• TROUBLESHOOTING QUICK-INDEX

DEFECT / SYMPTOM	LIKELY CAUSE	FIX
Dull / milky deposit	Temp or CD off the window; ratio off; CrO ₃ low	Lab-check ratio & CrO ₃ ; correct temp/CD to window
Burnt / rough at edges & tips	CD too high there; no robbers	Add robbers; adjust racking/CD
Bare/thin in recesses / bores	Poor throwing power (normal); missing conforming/internal anode	Add conforming/internal anode; re-rack
Poor adhesion / peeling	Bad/insufficient activation; not bare-active; contamination	Re-etch (anodic); verify cleaning & in-tank activation
Blistering / skip plating	High trivalent Cr ³⁺ ; contamination	Lab oxidize/treat per chemist; check anode condition
Coarse macrocracks	High stress; temp/CD/ratio drift	Lab review of bath parameters
Slow build / low rate	Low CD; cool bath; ratio off; tired anodes	Verify CD/temp/ratio; condition anodes
No plating at all	Wrong polarity; crusted/idle anodes; ratio far off; bad contact	Check polarity; condition anodes; lab check; re-rack
Undersize after grinding	Uneven plating (thin recess); ground through	Improve even coverage; allow more stock; re-plate
Grinding cracks / burn	Aggressive grinding; insufficient coolant	Light passes; correct wheel; ample coolant
Delayed cracking in service	Hydrogen embrittlement — bake skipped/late	Bake to ASTM B850; never skip on flagged parts
Visible mist over tank	Dead foam blanket; weak exhaust	Re-dose suppressant; fix exhaust; STOP if unsafe
High Cr in discharge	Treatment incomplete (reduce/precipitate)	Verify reduce-then-precipitate; do not discharge



TIP

Most hard-chrome defects trace to one of three roots: the **ratio/parameters** (lab), the **throwing-power setup** (anodes/robbers), or **activation/cleaning** (upstream). Name the root, not just the symptom.

— EVERY TERM, PLAIN AND SIMPLE

GLOSSARY

IF A WORD IN THIS MANUAL STUMPED YOU, IT'S DEFINED HERE

- **Action level (Cr VI):** 2.5 µg/m³ air; the trigger for monitoring, training, and medical surveillance under OSHA 1910.1026.
- **Ampere-hour (A·h):** amps × hours; the master measure of how much you've plated. Controls thickness.
- **Anode:** the positive electrode. In hard chrome it's **insoluble lead/lead-tin** — carries current, does *not* feed metal.
- **Cathode:** the negative electrode — your part — where chrome deposits.
- **Cathode efficiency:** percent of current that makes metal. Hard chrome ≈ **12–25%** (very low).
- **Chrome ulcer / “chrome hole”:** a deep, slow-healing skin ulcer from chromic-acid contact on broken skin.
- **Chromic acid:** CrO₃ dissolved in water; the hard-chrome bath. Contains **hexavalent (carcinogenic)** chromium.
- **Conforming anode:** a custom anode shaped to the part for even current/thickness.
- **Catalyst (sulfate/fluoride):** the trace additive that makes chrome deposit; target **CrO₃:SO₄ ≈ 100:1**.
- **Cr(VI) / hexavalent chromium:** the +6 form; a confirmed human carcinogen.
- **Cr(III) / trivalent chromium:** the +3 form; far less hazardous; the goal of waste reduction; also an emerging plating alternative.
- **Current density:** amps per unit area (A/dm² or ASF). Hard chrome ≈ **30–60+ A/dm²**.
- **Drag-out / drag-in:** solution clinging to a part leaving a tank (out) / contaminating the next tank (in).
- **Fume suppressant / foam blanket:** surfactant additive that suppresses Cr(VI) mist at the bath surface.
- **HRC / ksi UTS:** hardness (Rockwell C) / tensile strength; flag high-strength steel for the bake.

• GLOSSARY, CONTINUED

- **Hydrogen embrittlement:** hydrogen absorbed into hard steel causing delayed brittle cracking; relieved by baking.
- **Insoluble anode:** an anode that carries current without dissolving (lead/lead-tin here).
- **Macrocrack / microcrack:** coarse cracking (defect) vs. fine dense cracking (normal, desirable).
- **PEL (Cr VI):** Permissible Exposure Limit, **5 $\mu\text{g}/\text{m}^3$** (8-hr avg), OSHA 1910.1026.
- **Push-pull ventilation:** a tank exhaust system that blows air across the surface (push) into an exhaust slot (pull) to capture mist.
- **Reduce-then-precipitate:** $\text{Cr}^{6+} \rightarrow \text{Cr}^{3+}$ (acidic + reducing agent), then raise pH to drop out chromium sludge — the waste-treatment method.
- **Robber / thief:** scrap metal that steals excess current from edges so they don't over-build/burn.
- **Significant surface:** the surface that must meet thickness/finish spec; drives the amp-hour calc.
- **Throwing power:** ability to plate evenly into recesses; hard chrome's is **poor**.
- **Vickers hardness (HV):** hardness measure; hard chrome $\approx$ **800–1100 HV**.



REMEMBER

Of all these terms, four matter most on this line: **Cr(VI)** (the carcinogen you're managing), **amp-hours** (how you control thickness), **throwing power** (the weakness you fix with anodes/robbers), and **hydrogen embrittlement** (why the bake is mandatory). Master those four ideas and you understand hard chrome.



FUN FACT

The word "chromium" comes from the Greek *chroma*, meaning "color" — chromium compounds are famously vivid (the yellow chromic-acid bath, green trivalent solutions, orange dichromates). The metal named for color is, ironically, prized in plating for being nearly colorless and bright.

— TEMPLATE — PRINT ONE PER OPERATOR

OPERATOR TRAINING MATRIX

HOW A SUPERVISOR PROVES — ON PAPER — THAT EACH OPERATOR IS TRAINED BEFORE RUNNING SOLO

★

REMEMBER

An operator is “line-qualified” only when every station they run is signed. **Nobody works any station — especially the chrome tank, etch, strip, or rectifier — until the Cr(VI) Safety / PPE & SDS column is signed.** Safety qualification comes first, always, and on this line it’s a carcinogen gate.

Sign-off levels: **O** = Observed only · **A** = Assisted under supervision · **Q** = Qualified to run solo · **T** = Qualified to Train others.

#	STATION / SKILL	O/A/Q/T	TRAINER INIT & DATE	OPERATOR INIT & DATE	RE-CERT DUE
1	Cr(VI) Safety / PPE / SDS / hygiene (all stations)	_____	_____	_____	_____
2	Station 01 — Inspect / Mask / Rack	_____	_____	_____	_____
3	Station 02 — Clean / Degrease + water-break	_____	_____	_____	_____
4	Station 03 — Rinse + conductivity	_____	_____	_____	_____
5	Station 04 — Anodic Etch / Activation	_____	_____	_____	_____
6	Station 05 — Hard Chrome Plating operation	_____	_____	_____	_____
7	Thickness control by ampere-hours	_____	_____	_____	_____
8	Conforming anodes / robbers / throwing power	_____	_____	_____	_____
9	Station 06 — Drag-out / Rinse + chrome recovery	_____	_____	_____	_____
10	Station 07 — H-Embrittlement Bake (ASTM B850)	_____	_____	_____	_____
11	Station 08 — Grind / Finish to size	_____	_____	_____	_____
12	Station 09 — Strip / Rework	_____	_____	_____	_____
13	Station 10 — Dry / Final Inspection	_____	_____	_____	_____
14	LOTO on rectifier	_____	_____	_____	_____
15	Emergency response — eyewash, shower, spill	_____	_____	_____	_____
16	Cr(VI) waste treatment & wastewater/air controls	_____	_____	_____	_____
17	Troubleshooting — symptom recognition	_____	_____	_____	_____

Operator name: _____ Hire/start date: _____

Supervisor name: _____ Review cycle: ☐ Annual ☐ Other: _____

⚠

HEADS-UP

Safety rows (1, 14, 15, 16) are not optional and not “fill in later.” An operator may not work *any* station until Cr(VI) PPE/SDS/hygiene, LOTO, emergency response, and waste-control awareness are signed. On a carcinogen line, safety competency absolutely precedes production competency.

PART FOUR — COMPLETION TEST, ANSWER KEY & CERTIFICATE

20 QUESTIONS • PASSING SCORE 80% (16/20)

HARD CHROME COMPLETION TEST**COVERS EVERY STATION PLUS SAFETY — ANSWER IN YOUR OWN WORDS WHERE ASKED****REMEMBER**

Passing score: 80% (16 of 20 correct). Any **safety** question answered wrong (**Q3, Q4, Q9, Q15, Q19**) is an automatic fail of the safety gate — re-teach and re-test before sign-off, regardless of total score. The Answer Key follows — no peeking until you're done!

Name: _____ Date: _____ Score: _____ / 20 **Safety-gate Qs (all must be correct): 3, 4, 9, 15, 19**

Q1. Hard chrome differs from decorative chrome mainly because hard chrome is:

- A. Plated thick, directly on steel, and usually ground to size B. Thinner and plated over nickel C. Always trivalent D. Never used on steel

Q2. The hard chrome bath is made mainly from:

- A. Zinc chloride B. Nickel sulfate C. Chromic acid (CrO_3) plus a catalyst D. Sodium cyanide

Q3. [SAFETY GATE] The chromium in the hard chrome bath is hexavalent — Cr(VI) . What is the most important fact about it?

- A. It is harmless B. It is a confirmed human carcinogen C. It is flammable D. It is a nutrient supplement

Q4. (Short answer) [SAFETY GATE] Name the OSHA Cr(VI) **PEL** and **action level** (with units), and list **three** controls your shop uses to keep chromic-acid mist out of your lungs.

Q5. The classic catalyst ratio in a hard chrome bath is:

- A. $\text{CrO}_3 : \text{SO}_4 \approx 1:1$ B. No catalyst is used C. $\text{CrO}_3 : \text{SO}_4 \approx 10:1$ D. $\text{CrO}_3 : \text{SO}_4 \approx 100:1$

Q6. Hard chrome's cathode efficiency is approximately:

- A. 95–98% B. 60–80% C. 12–25% D. 100%

Q7. (Short answer) Hard chrome uses **insoluble lead/lead-tin anodes**. Explain how that's different from zinc anodes, and where the chromium actually comes from.

Q8. Typical hard chrome bath temperature and cathode current density are about:

- A. $\sim 50\text{--}60^\circ\text{C}$; $\sim 30\text{--}60\text{ A/dm}^2$ B. Room temp; $1\text{--}6\text{ A/dm}^2$ C. Below freezing; 1 A/dm^2 D. 95°C ; 0.5 A/dm^2

Q9. (Short answer) [SAFETY GATE] Why does hard chrome create a serious **hydrogen-embrittlement** risk, and what is the mandatory fix for high-strength steel (give the typical temperature and minimum time)?

Q10. Hard chrome's **throwing power** is poor. The three mechanical tools used to get even coverage are:

- A. Brighteners, carriers, levelers B. Conforming anodes, robbers/thieves, and careful racking C. Higher pH, more current, hotter bath D. Cyanide, ammonia, boric acid

Q11. (Short answer) What is an **ampere-hour**, and why is amp-hour control the core operator skill for hitting a target thickness in hard chrome?

Q12. A part keeps coming out **bare/thin inside a deep bore**. The most likely fix is:

A. More current overall B. A hotter bath C. More brightener D. An internal/conforming anode inside the bore

Q13. Typical hard chrome deposit hardness is:

A. 100–200 HV B. 400–500 HV C. 800–1100 HV D. 2000–3000 HV

Q14. A good hard chrome deposit shows a fine, dense **microcrack** pattern. This is:

A. Normal and desirable for wear B. A defect — reject the part C. A sign of contamination D. Caused by the bake

Q15. (Short answer) [SAFETY GATE] Describe the **reduce-then-precipitate** method for treating Cr(VI) wastewater (what you do in each step and what the goal is), and state one place Cr(VI) water must **never** go.

Q16. Hard chrome is plated **oversize** and then:

A. Painted B. Ground/finished to final dimension C. Dipped in nickel D. Left as-is

Q17. (Short answer) Why must the hydrogen-embrittlement **bake** come **before** final grinding, and why must it start **promptly** after plating?

Q18. Why must the **foam blanket / fume suppressant** and **exhaust** keep working for the **entire** plating run, not just the start?

A. To keep the part warm B. To speed up plating C. It doesn't matter D. Because the run lasts hours and Cr(VI) mist exposure continues the whole time



SAFETY SECTION — MUST BE CORRECT TO CERTIFY

Questions 3, 4, 9, 15, and 19 cover hazards that can permanently harm you or others. A miss on any safety item means re-teach and re-test before sign-off.

Q19. (Short answer / Safety) [SAFETY GATE] List **four** pieces of PPE you must wear at the hard chrome tank, and name the procedure required before any maintenance on the high-amperage **rectifier**.

Q20. (Short answer) Name **two** ways hard (functional) chrome is fundamentally different from acid zinc plating.

End of test. Hand to your trainer for grading.

ANSWER KEY

SHORT-ANSWER RESPONSES DON'T NEED TO MATCH WORD-FOR-WORD — LOOK FOR THE KEY CONCEPTS IN BOLD



HEADS-UP

Keep this page out of the trainee's copy. Questions **3, 4, 9, 15, and 19** are safety-critical — any wrong answer there requires re-training on that topic before sign-off, regardless of total score.

- Q1. A** — Plated thick, directly on steel, and usually ground to size (functional, not decorative).
- Q2. C** — Chromic acid (CrO_3) plus a catalyst.
- Q3. B** — It is a **confirmed human carcinogen**.
- Q4. PEL = 5 $\mu\text{g}/\text{m}^3$, action level = 2.5 $\mu\text{g}/\text{m}^3$** (8-hr averages, OSHA 29 CFR 1910.1026). Three controls (any three): **fume suppressant/foam blanket, local exhaust / push-pull ventilation, fume scrubber, respiratory protection, hygiene/no eating-smoking**. (Engineering controls before PPE.)
- Q5. D** — $\text{CrO}_3 : \text{SO}_4 \approx 100:1$.
- Q6. C** — **12–25%** (very low).
- Q7.** Zinc anodes are **soluble** — they dissolve to feed zinc into the bath. Hard chrome anodes are **insoluble lead/lead-tin** — they carry current and keep the bath oxidized but do **not** dissolve. The chromium comes from the **chromic acid (CrO_3) maintained as a chemical** in the bath.
- Q8. A** — $\sim 50\text{--}60^\circ\text{C}$; $\sim 30\text{--}60\text{+ A}/\text{dm}^2$ (300–600+ ASF).
- Q9.** Low efficiency ($\sim 12\text{--}25\%$) generates **a lot of hydrogen** at the part, which **soaks into high-strength steel and embrittles it** (delayed brittle cracking under load). Fix: **bake $\sim 375^\circ\text{F}$ (190°C) for ≥ 8 hours**, started **promptly** after plating (ASTM B850). Mandatory on high-strength steel.
- Q10. B** — Conforming anodes, robbers/thieves, and careful racking.
- Q11.** An **ampere-hour** = amps $\times$ hours of plating (total charge passed). Because thickness depends on charge $\times$ area $\times$ efficiency (Faraday + $\sim 12\text{--}25\%$ efficiency), you **count amp-hours** to hit a target thickness — “amp-hours make microns.” (Verify with a thickness check.)
- Q12. D** — An internal/conforming anode inside the bore (poor throwing power means outside anodes can't reach the ID).
- Q13. C** — 800–1100 HV.
- Q14. A** — Normal and desirable; microcracks hold lubricant and relieve stress. (Coarse *macrocracks* are the defect.)
- Q15. Reduce $\text{Cr}^{6+} \rightarrow \text{Cr}^{3+}$** with a reducing agent at **acidic/low pH**, then **raise the pH** to **precipitate** the chromium as insoluble hydroxide sludge and **filter/settle** it out; goal is to convert the carcinogenic hexavalent form to the far-less-hazardous trivalent form and remove it. **Cr(VI) water must never** go down a floor drain / to the sewer untreated.
- Q16. B** — Ground/finished to final dimension.
- Q17.** Bake **before grinding** so the part is hydrogen-relieved and metallurgically settled when machined to its final critical dimension; bake **promptly** because hydrogen can crack a high-strength part *before* baking if you wait too long.
- Q18. D** — The run lasts hours, so **Cr(VI) mist exposure** continues the whole time; controls must work throughout.
- Q19. PPE (any four):** sealed chemical splash goggles (not safety glasses), face shield, chemical-resistant gloves, chemical apron/suit, chemical-resistant boots, respirator per the **Cr(VI) program**. **Rectifier procedure: LOTO (Lockout/Tagout)** before any maintenance.
- Q20.** Any two of: hard chrome is **functional/thick and ground to size** (zinc is thin sacrificial); **insoluble lead anodes** vs. soluble zinc anodes; **very low efficiency ($\sim 12\text{--}25\%$)** vs. $\sim 95\text{--}98\%$; **very high CD ($\sim 30\text{--}60\text{+ A}/\text{dm}^2$)** vs. $\sim 1\text{--}6$; **hot bath ($\sim 50\text{--}60^\circ\text{C}$)** vs. ambient; **Cr(VI) carcinogen** chemistry; **poor throwing power needing conforming anodes/robbers**; **mandatory embrittlement bake**; **microcracked deposit**.

Answer distribution (MC): A=3 · B=3 · C=3 · D=3



REMEMBER

A score of 16/20 passes — but every safety question (3, 4, 9, 15, 19) must be correct to certify. Miss a safety item → re-teach and re-test before sign-off. On a **Cr(VI)** line, we don't gamble with people.

PRINT, FILL IN, SIGN



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CERTIFICATE OF COMPLETION
HARD CHROME PLATING TRAINING PROGRAM

This certifies that

OPERATOR NAME

has successfully completed the **Hard Chrome (Functional Chromium) Plating** training course — covering the full functional hard-chrome workflow (Inspect/Mask/Rack, Clean, Rinse, Anodic Etch/Activation, Plate, Drag-out/Rinse, Hydrogen-Embrittlement Bake, Grind/Finish, Strip/Rework, and Final Inspection), chromic-acid bath chemistry and the 100:1 catalyst ratio, insoluble-anode operation, thickness control by ampere-hours, conforming anodes/robbars and throwing power, hydrogen-embrittlement control and baking, and — above all — the **hexavalent chromium (Cr VI) safety practices** required of every operator (mist suppression, ventilation, respiratory protection, hygiene, spill response, and Cr(VI) waste treatment) — and has passed the Completion Test with a score of _____ / 20, including all required safety competencies.

This operator is hereby cleared for **independent work** at the certified stations listed on the Operator Training Matrix.

Date of completion: _____ Re-certification due: _____

TRAINEE SIGNATURE

CERTIFYING TRAINER

SUPERVISOR

Training reference only. Verify all parameters against your chemistry supplier's TDS, customer specifications, and applicable regulatory requirements before production use. **Hexavalent chromium is a confirmed human carcinogen** — always consult current SDS and comply with OSHA 29 CFR 1910.1026, EPA, REACH, and local regulations.

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